

# Banking Question Bank - Set 1

394 questions with verified answer keys

## DIRECTIONS (Q. 1-3)

In each of these questions, two equations (I) and (II) are given. You have to solve both the equations and give answer.

1. I.  $11x + 64/x = 54$

II.  $12y^2 + 40y + 17 = 0$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (a) if  $x > y$**

2. I.  $84x^2 + 115x + 26 = 0$

II.  $27y^2 + 36y - 15 = 0$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (d) if  $x < y$**

3. I.  $24x^2 + 25x - 11 = 0$

II.  $45y^2 + 36y + 7 = 0$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (a) if  $x > y$**

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4. A shopkeeper sells two pens, he sold 1 pen at profit and other pen at loss. SP of each of the two pens is Rs.300 and profit percentage on 1 pen is equal to loss percentage on other. If overall loss of shopkeeper is 6.25%, then find difference between cost price of both pen.

- (a) Rs.350
- (b) Rs.100
- (c) Rs.240
- (d) Rs.160
- (e) Rs.300

**Answer (c) Rs.240**

5. In a class percentage of students who passed the exam is 60% and number of boys & girls who passed the exam is same. If boys who failed the exam are 200% more than girls who failed in exam then find the percentage of girls who failed out of total students

- (a) 9%
- (b) 13%
- (c) 10%
- (d) 12%
- (e) 15%

**Answer (d) 12%**

**DIRECTIONS (Q. 6-8)**

Study the following arrangement carefully and answer the questions given below: D F 3 # R N 8 A @ Y 4 M ★ W P 6 H U 9 1 K 2 E

6. How many such consonants are there in the above arrangement each of which is immediately preceded by a symbol and immediately followed by a letter?
- (a) None
  - (b) One
  - (c) Two
  - (d) Three
  - (e) None of these

**Answer (b) One**

7. If all numbers are dropped from the above arrangement, which of the following will be the fourteenth from the left end?
- (a) I
  - (b) P
  - (c) W
  - (d) U
  - (e) None of these

**Answer (a) I**

8. How many such numbers are there in the above arrangement each of which is immediately preceded by a vowel?
- (a) None
  - (b) One
  - (c) Two
  - (d) Three
  - (e) None of these

**Answer (b) One**

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9. A container contain 96 litre mixture of milk and water out of which 35% is water. 20 lit of mixture is taken out and 10 litre milk & 16 litre water are added in remaining mixture. Find the difference between quantity of milk and water in the final mixture after replacement.
- (a) 15.4 lit
  - (b) 16.8 lit
  - (c) 18.2 lit
  - (d) 21.4 lit
  - (e) 19.8 lit

**Answer (a) 15.4 lit**

**DIRECTIONS (Q. 10-11)**

Select the phrase/connector (STARTERS) from the given three options which can be used to form a single sentence from the two sentences given below, implying the same meaning as expressed in the statement sentences.

10. China will step up efforts to contain the coronavirus outbreak ahead of the Lunar New Year holidays. 17 new cases of the virus have been confirmed and there is increasing possibility of spread of virus to other countries. (i) In lieu of 17 new cases... (ii) Owing to the 17 new cases.... (iii) Taking into consideration China's efforts to....
- (a) Only (i)
  - (b) Only (ii)
  - (c) Only (iii)
  - (d) Both (i) and (ii)

(e) None of the above

**Answer (a) Only (i)**

11. Doubts were raised about the efficacy of the Right To Information Act in Haryana. Only 35% penalty has been recovered since the setting up of the Haryana State Information Commission in 2006. (i) Citing the modus operandi of RTI..... (ii) Stating the facts regarding cases under RTI.... (iii) In what raises doubts about the efficacy.....

- (a) Only (i)
- (b) Only (ii)
- (c) Only (iii)
- (d) Both (iii) and (i)
- (e) None of the above

**Answer (c) Only (iii)**

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12. Azhar Ali plays for which of the following team?

- (a) Durham County Cricket Club
- (b) Lancashire County Cricket Club
- (c) Worcestershire County Cricket Club
- (d) Nottinghamshire County Cricket Club
- (e) Kent County Cricket Club

**Answer (d) Nottinghamshire County Cricket Club**

13. Which of the following combinations is true?

- (a) Chris Gayle – Somerset County Cricket Club - Bangladesh
- (b) Lasith Malinga – Nottinghamshire County Cricket Club – Sri Lanka
- (c) Mushfiquir Rahim – Glamorgan County Cricket Club – Australia
- (d) Ross Taylor – Essex County Cricket Club - New Zealand
- (e) None of these

**Answer (d) Ross Taylor – Essex County Cricket Club - New Zealand**

14. Bala working in which of the following country?

- (a) UAE
- (b) Saudi Arabia
- (c) Egypt
- (d) Yemen
- (e) Palestine

**Answer (d) Yemen**

15. Daniel Ricciardowon which of the following grand prix?

- (a) Russian Grand Prix
- (b) Australian Grand Prix
- (c) Chinese Grand Prix
- (d) Canadian Grand Prix
- (e) Bahrain Grand Prix

**Answer (b) Australian Grand Prix**

16. Which month Sebastian Vettel was born?

- (a) February
- (b) April
- (c) August
- (d) November
- (e) Cannot be determined

**Answer (c) August**

17. Which of the following product does Vishal order?

- (a) Books
- (b) Pen drive
- (c) Mobile
- (d) Shirt
- (e) Watch

**Answer (a) Books**

18. If 'Laptop' is related to 'Paytm', 'Pen drive' is related to 'Amazon', then which following is 'Watch' related to?

- (a) Flipkart
- (b) Jabong
- (c) Snapdeal
- (d) Ebay
- (e) None of these

**Answer (c) Snapdeal**

19. The person who saved money on August, Saved \_\_\_\_\_

- (a) 40
- (b) 80
- (c) 500
- (d) 2500
- (e) 1600

**Answer (a) 40**

20. Four among the following form a group in a certain way. Which of the following does not belong to Group?

- (a) A
- (b) F
- (c) C
- (d) B
- (e) D

**Answer (e) D**

21. Which of the following Bank listed on 3rd place from the top?

- (a) Andhra Bank
- (b) Dena Bank
- (c) Syndicate Bank
- (d) Canara Bank
- (e) Bank of Baroda

**Answer (c) Syndicate Bank**

22. If 'Punjab National Bank' is related to 'Canara Bank', 'Syndicate Bank' is related to 'Vijaya Bank', then which following Bank is 'Dena Bank' related to?

- (a) Vijaya Bank
- (b) Canara Bank
- (c) Andhra Bank
- (d) Bank of Baroda
- (e) Punjab National Bank

**Answer (d) Bank of Baroda**

23. Which of the following month Jadhav and Maran wrote the exam?

- (a) October
- (b) April
- (c) March
- (d) June
- (e) August

**Answer (b) April**

24. Which of the following student not wrote the exam on the day 9th?

- (a) Gowtham
- (b) Dinesh
- (c) Gopi
- (d) Harish
- (e) Naren

**Answer (a) Gowtham**

25. How many people live between the one who working in Punjab National Bank and the one who working in Union Bank of India?

- (a) Two
- (b) Three
- (c) Four
- (d) Five
- (e) Six

**Answer (c) Four**

26. If 'Bank of Baroda' is related to 'Rajasthan', 'Canara Bank' is related to 'Punjab', then which following is 'Punjab National Bank' related to?

- (a) Karnataka
- (b) Tamil Nadu
- (c) Kerala
- (d) Gujarat
- (e) Haryana

**Answer (b) Tamil Nadu**

27. Who lives in the sixth floor?

- (a) The one who is from Infosys
- (b) The one who is from HCL
- (c) The one who is from Mindtree
- (d) The one who is from TCS
- (e) The one who is from Mphasis

**Answer (a) The one who is from Infosys**

28. How many persons using Bus as a transport?

- (a) One
- (b) Two
- (c) Three
- (d) Four
- (e) Five

**Answer (c) Three**

**29.** Who lives in the Third floor?

- (a) Dharish
- (b) The one who likes Basketball
- (c) The one who is from China
- (d) Madhavan
- (e) None of these

**Answer (c) The one who is from China**

**30.** Sanjay buys which of the following magazines?

- (a) Wisdom
- (b) India Today
- (c) Digit
- (d) Business Today
- (e) Business world

**Answer (d) Business Today**

**31.** How many persons live between Ajay and Hemant in the third flat?

- (a) One
- (b) Two
- (c) Three
- (d) None
- (e) Cannot be determined

**Answer (b) Two**

**32.** Which of the following mobile does Pavithra have?

- (a) Samsung
- (b) Oppo
- (c) Motorola
- (d) Sony
- (e) Vivo

**Answer (e) Vivo**

**33.** Mahima lives on which of the following floor?

- (a) 8th floor
- (b) 6th floor
- (c) 4th floor
- (d) 2nd floor
- (e) 3rd floor

**Answer (a) 8th floor**

**34.** Divya likes which of the following Biryani?

- (a) AmburBiryani
- (b) Afghani Biryani
- (c) Bombay Biryani
- (d) KacchiBiryani
- (e) Kolkata Biryani

**Answer (e) Kolkata Biryani**

**35.** Shankar likes which of the following juice?

- (a) Apple juice
- (b) Pomegranate juice
- (c) Orange juice
- (d) Mango juice
- (e) Pineapple juice

**Answer (e) Pineapple juice**

36. The person who selected in SSC CGL exam is working in which of the following state?

- (a) Odisha
- (b) Punjab
- (c) Haryana
- (d) Both Odisha and Haryana
- (e) Gujarat

**Answer (d) Both Odisha and Haryana**

37. Who among the following working in Gujarat?

- (a) R
- (b) P
- (c) O
- (d) M
- (e) None of these

**Answer (a) R**

38. Vishnu have which of the following bike?

- (a) Apache bike
- (b) Glamour bike
- (c) Hornet bike
- (d) Pulsar bike
- (e) None of these

**Answer (b) Glamour bike**

**DIRECTIONS (Q. 39)**

Study the following information carefully and answer the given questions. A number arrangement machine arranges two digit numbers into a typical manner. Each step gives output taking input from the previous step. The following is an illustration of Input and rearrangement. Using the illustration answer the question given below.  
Input: 3 4 5 3 2 4 2 1 3 1 2 4 Step I 3 8 5 9 8 8 Step II 2 0 2 1 Step III 8 9 Step IV 17 Input:

39. Which of the following represent the value, the second digit of the third number in step I divided by the second digit of the second number in step II?

- (a) 4
- (b) 1
- (c) 6
- (d) 3
- (e) 2

**Answer (d) 3**

**DIRECTIONS (Q. 40)**

Study the following information carefully and answer the given questions. A number arrangement machine arranges two digit numbers into a typical manner. Each step gives output taking input from the previous step. The following is an illustration of Input and rearrangement. Using the illustration answer the question given below.  
136 Input: 3 5 8 3 4 6 4 5 2 1 2 4 Step I 8 9 9 5 8 8 Step II 5 0 4 4 Step III 1 -4 Step IV -3 Input:

40. What is the difference of both numbers obtained in step II?

- (a) 5
- (b) 10
- (c) 8
- (d) 6
- (e) 9

**Answer (d) 6**

**DIRECTIONS (Q. 41)**

Study the following information carefully and answer the given questions. A number arrangement machine arranges two digit numbers into a typical manner. Each step gives output taking input from the previous step. The following is an illustration of Input and rearrangement. Using the illustration answer the question given below.  
Input: 8 7 5 3 9 6 1 4 2 3 3 4 Step I 4 6 2 1 5 3 Step II 1 0 1 1 139 Step III 1 2 Step IV 2 Input:

41. If both numbers in the step II is divided by 2, then what will be the sum of both numbers?

- (a) 24
- (b) 12
- (c) 10
- (d) 36
- (e) 48

**Answer (b) 12**

**DIRECTIONS (Q. 42)**

Study the following information carefully and answer the given questions. A number arrangement machine arranges two digit numbers into a typical manner. Each step gives output taking input from the previous step. The following is an illustration of Input and rearrangement. Using the illustration answer the question given below.  
Input: 1 6 8 4 3 1 8 9 2 2 3 2 Step I 1 2 4 2 4 3 Step II 1 6 1 2 Step III 32 24 Step IV 8 Input:

42. If each digit in the step I is multiply with 2 and then added, then what will be the sum of all numbers in step I?

- (a) 28
- (b) 42
- (c) 34
- (d) 52
- (e) 20

**Answer (c) 34**

**DIRECTIONS (Q. 43)**

Study the following information carefully and answer the given questions. A number arrangement machine arranges two digit numbers into a typical manner. Each step gives output taking input from the previous step. The following is an illustration of Input and rearrangement. Using the illustration answer the question given below.  
Input: 6 5 7 3 5 1 1 6 3 9 7 5 Step I 3 6 5 6 3 9 2 5 7 Step II 1 1 1 4 2 1 Step III 0 2 0 4 Step IV 2 4 Step V 2 Input:

43. If all the three numbers in the step II is added, then what will be the final sum?

- (a) 45
- (b) 39
- (c) 63
- (d) 58
- (e) 53

**Answer (e) 53**

**DIRECTIONS (Q. 44)**

Study the following information carefully and answer the given questions. A number arrangement machine arranges two digit numbers into a typical manner. Each step gives output taking input from the previous step. The following is an illustration of Input and rearrangement. Using the illustration answer the question given below.  
Input: 2 3 9 4 2 9 7 6 4 7 3 8 Step I 1 1 5 1 6 8 1 5 9 Step II 0 5 4 8 4 5 Step III 0 8 1 8 Step IV 1 0 Step V 1 Input: 7 5 8 3 2 6 9 5 2 4 1 8



44. If the value '3' is subtracted from the final output, then what will be the resultant value?

- (a) 9
- (b) 13
- (c) 7
- (d) 8
- (e) 5

**Answer (c) 7**

**DIRECTIONS (Q. 45)**

Study the following information carefully and answer the given questions. A number arrangement machine arranges two digit numbers into a typical manner. Each step gives output taking input from the previous step. The following is an illustration of Input and rearrangement. Using the illustration answer the question given below.

Input: 7 5 4 6 1 7 9 3 1 8 3 9 Step I 4 9 5 3 2 6 8 1 9 Step II 3 6 2 2 3 6 Step III 0 8 1 4 Step IV 8 5 Step V 40  
Input:

45. What is the sum of all three numbers obtained in step II?

- (a) 96
- (b) 100
- (c) 98
- (d) 106
- (e) 88

**Answer (c) 98**

46. What may be the possible code for 'so calculator' in the given code language?

- (a) 04&TC, 20\*RA
- (b) 21!QB, 16&UG
- (c) 08#XL, 08#KD
- (d) 21!QB, 09!ZW
- (e) 27\*JT, 09!ZW

**Answer (a) 04&TC, 20\*RA**

47. What may be the possible code for 'have interesting Truthness' in the given code language?

- (a) 08!IR, 08&KL, 27\*JT
- (b) 33&LP, 08&KL, 16+BQ
- (c) 09!ZW, 15+VY, 27\*JT
- (d) 33&LP, 08&KL, 27\*JT
- (e) 09!ZW, 08&KL, 27\*JT

**Answer (d) 33&LP, 08&KL, 27\*JT**

48. What may be the possible word for '03\*Q, 16#H, 14#R' in the given code language?

- (a) repeat welled good
- (b) fault crow repeat
- (c) once tonner repeat
- (d) the good tonner
- (e) None of these

**Answer (c) once tonner repeat**

49. What may be the possible code for 'answer hiring' in the given code language?

- (a) 21%R, 12#O
- (b) 19\*Q, %L18
- (c) 09@K, \*N09
- (d) 04\$D, @K17
- (e) 21%R, \$J21

**Answer (b) 19\*Q, %L18**

50. What may be the possible word for “30!T, 20%M, 72\*U” in the given code language?

- (a) pre named rush
- (b) to book societies
- (c) repayment for cousins
- (d) book their counters
- (e) None of these

**Answer (d) book their counters**

51. What is the code for ‘viewed’ in the given code language?

- (a) 20%M
- (b) 12\*F
- (c) 42@F
- (d) 12\$T
- (e) None of these

**Answer (c) 42@F**

52. What may be the possible code for ‘coach pre’ in the given code language?

- (a) 12\$T, 12#F
- (b) 12\*F, 30!J
- (c) 20%M, 06&U
- (d) 42@F, 12#F
- (e) 20%M, 56%F

**Answer (b) 12\*F, 30!J**

53. What may be the possible word for ‘02@C, 23#X, 20!U ’ in the given code language?

- (a) Modem catch jewel
- (b) Like took traffic
- (c) Late program difficult
- (d) wrong batches traffic
- (e) None of these

**Answer (d) wrong batches traffic**

54. What is the possible for ‘03\*D, 10!K’ in the given code language?

- (a) catch jewel
- (b) Maps entry
- (c) difficult Like
- (d) entry batches
- (e) None of these

**Answer (a) catch jewel**

55. What may be the possible code for ‘Maps reaction’ in the given code language?

- (a) 23#X, 15@P
- (b) 16!Q, 15@P
- (c) 23#X, 12#M
- (d) 20%U, 20!U
- (e) 13%N, 18\*S

**Answer (e) 13%N, 18\*S**

56. What is the code for ‘Sole trader power’ in the given code language?

- (a) #R18, &P22, %S20
- (b) #Q16, &U20, %T19
- (c) #S14, &T18, %P21
- (d) #M10, &O15, %T17
- (e) None of these

**Answer (d) #M10, &O15, %T17**

57. What is the code for 'That' in the given code language?

- (a) 19\*I
- (b) 09!D
- (c) 21&J
- (d) 23%K
- (e) None of these

**Answer (b) 09!D**

58. What is the code for 'dancing back benchers' in the given code language?

- (a) 15&G,25\$L
- (b) 19%I, 27@M
- (c) 13#F, 07#C
- (d) 17@H, 27@M
- (e) 19%I, 11%E

**Answer (a) 15&G,25\$L**

59. What may be the possible word for "27%M, 17@H, 27@M" in the given code language?

- (a) Childrens mega project abbreviations
- (b) Megaproject backbenchers priced
- (c) to magazine Pages
- (d) Spontaneously magazine abbreviations
- (e) None of these

**Answer (d) Spontaneously magazine abbreviations**

60. What may be the possible code for 'readjusts priced' in the given code language?

- (a) 09!D, 13#F
- (b) 23%K, 15&G
- (c) 21&J, 17@H
- (d) 23%K, 19%I
- (e) 19\*I, 13#F

**Answer (e) 19\*I, 13#F**

61. What is the code for 'lorry' in the given code language?

- (a) 22+B
- (b) 12#L
- (c) 02\*V
- (d) 11&M
- (e) None of these

**Answer (c) 02\*V**

62. What is the possible word for '12#L, 20!D' in the given code language?

- (a) Auto Lightning
- (b) hold layer
- (c) layer Auto
- (d) roundup people
- (e) None of these

**Answer (a) Auto Lightning**

63. What may be the possible code for 'house Lightning' in the given code language?

- (a) 22\*B, 23!A
- (b) 08&P, 13#K
- (c) 19\*E, 22\$B
- (d) 13&K, 23!A
- (e) 22\*B, 20!D

**Answer (e) 22\*B, 20!D**

64. What is the code for 'Animals people' in the given code language?

- (a) 13&K, 20!D
- (b) 08&P, 22\$B
- (c) 22\*B, 22\$B
- (d) 19\*E, 23!A
- (e) 08&P, 13#K

**Answer (b) 08&P, 22\$B**

65. What is the code for 'started' in the given code language?

- (a) 08\$S
- (b) 02#X
- (c) 08\$P
- (d) 04+T
- (e) None of these

**Answer (d) 04+T**

66. What may be the possible code for 'arranging supermarket' in the given code language?

- (a) 08\$R, 04+T
- (b) 06%R, 08\$P
- (c) 06%U, 08\$S
- (d) 04+T, 02#X
- (e) 04+T, 08\$S

**Answer (b) 06%R, 08\$P**

67. What is the possible word for "06%T, 04+X" in the given code language?

- (a) renovating plans
- (b) plans sales
- (c) downtown in
- (d) renovating worked
- (e) Sukumar are

**Answer (e) Sukumar are**

68. What is the code for 'downtown mountains' in the given code language?

- (a) 04+S, 08\$R
- (b) 02#V, 04+T
- (c) 04+X, 04+T
- (d) 08\$Q, 06%R
- (e) 02#V, 06%U

**Answer (a) 04+S, 08\$R**

69. What is the code for '06#A, 42#E' in the given code language?

- (a) big warehouse
- (b) warehouse have
- (c) Ishwarya is
- (d) cute warehouse
- (e) big friends

**Answer (e) big friends**

70. What may be the possible word for '42#E, 56@L, 06#G' in the given code language?

- (a) Teachers cute friends
- (b) friends motivate him
- (c) have motivate teachers
- (d) you motivate him
- (e) None of these

**Answer (b) friends motivate him**

71. What may be the possible code for 'Teachers have' in the given code language?

- (a) 56@L, 12@B
- (b) 06#G, 56%H
- (c) 56#S, 12@G
- (d) 12#V, 12@B
- (e) None of these

**Answer (c) 56#S, 12@G**

72. What is the code for 'Ishwarya must well' in the given code language?

- (a) 12@B, 12#G, 56@L
- (b) 42#E, 12#L, 56#S
- (c) 12@G, 06#A, 06#G
- (d) 56%H, 12#L, 12#V
- (e) None of these

**Answer (b) 42#E, 12#L, 56#S**

73. What is the word for '07#K, 13&H' in the given code language?

- (a) Above upto
- (b) Army hack
- (c) recover fires
- (d) embassy tabs
- (e) hack embassy

**Answer (e) hack embassy**

74. What may be the possible word for '15@L, "13&U, 11\$Q' in the given code language?

- (a) Unable tabs refer
- (b) refer recover eggs
- (c) Hit remember above
- (d) impaired recover number
- (e) None of these

**Answer (d) impaired recover number**

75. What is the code for 'hack unimportant eggs' in the given code language?

- (a) 07#K, 21!X, 07#H
- (b) 09\*D, 09\*K, 05+K
- (c) 07#W, 15@U, 15@L
- (d) 13&U, 11\$Q, 11\$X
- (e) None of these

**Answer (a) 07#K, 21!X, 07#H**

76. What is the code for 'Children' in the given code language?

- (a) 09%G
- (b) 09#V
- (c) 16#O
- (d) 36%D
- (e) None of these

**Answer (d) 36%D**

77. What may be the possible code for 'Union Minister' in the given code language?

- (a) 25%J, 01#P
- (b) 49%H, 09%D
- (c) 04%I, 09%B
- (d) 04#V, 01#P
- (e) 04#V, 25%N

**Answer (e) 04#V, 25%N**

78. What is the possible word for “25%B,09%D” in the given code language?

- (a) Assam emergency
- (b) ambulance chief
- (c) Minister service
- (d) provide service
- (e) emergency minister

**Answer (b) ambulance chief**

79. What is the code for ‘provide fund’ in the given code language?

- (a) 16#Q, 09%G
- (b) 25%B, 09#V
- (c) 16#Q, 36%D
- (d) 36%F, 16#O
- (e) None of these

**Answer (a) 16#Q, 09%G**

80. I.  $2x^2 - 15\sqrt{3}x + 84 = 0$

II.  $3y^2 - 10\sqrt{3}y + 9 = 0$

- (a)  $x > y$
- (b)  $x < y$
- (c)  $x \geq y$
- (d)  $x \leq y$
- (e)  $x = y$  or relation cannot be established

**Answer (a)  $x > y$**

81. I.  $2x^2 - (8 + \sqrt{3})x + 4\sqrt{3} = 0$

II.  $3y^2 - (6 + 2\sqrt{3})y + 4\sqrt{3} = 0$

- (a)  $x > y$
- (b)  $x < y$
- (c)  $x \geq y$
- (d)  $x \leq y$
- (e)  $x = y$  or relation cannot be established

**Answer (b)  $x < y$**

82. I.  $x^2 - (2 + \sqrt{5})x + 2\sqrt{5} = 0$

II.  $2y^2 - (6 + 3\sqrt{5})y + 9\sqrt{5} = 0$

- (a)  $x > y$
- (b)  $x < y$
- (c)  $x \geq y$
- (d)  $x \leq y$
- (e)  $x = y$  or relation cannot be established

**Answer (b)  $x < y$**

83. I.  $3x^2 + 5\sqrt{2}x - 24 = 0$

II.  $y^2 - 6\sqrt{2}y + 16 = 0$

- (a)  $x > y$
- (b)  $x < y$
- (c)  $x \geq y$
- (d)  $x \leq y$
- (e)  $x = y$  or relation cannot be established

**Answer (b)  $x < y$**

84. I.  $3x^2 - 14x + 8 = 0$   
II.  $3y^2 - 20y + 12 = 0$
- (a)  $x > y$
  - (b)  $x < y$
  - (c)  $x \geq y$
  - (d)  $x \leq y$
  - (e)  $x = y$  or relation cannot be established

**Answer (a)  $x > y$**

85. I.  $12x^2 + 25x + 12 = 0$   
II.  $3y^2 + 22y + 24 = 0$
- (a)  $x > y$
  - (b)  $x < y$
  - (c)  $x \geq y$
  - (d)  $x \leq y$
  - (e)  $x = y$  or relation cannot be established

**Answer (a)  $x > y$**

**DIRECTIONS (Q. 86)**

In each of these questions two equations numbered I and II are given. You have to solve both the equations and give answer.

86. I.  $x^2 - 32x + 256 = 0$   
II.  $y^2 - 33y + 272 = 0$
- (a)  $x < y$
  - (b)  $x \leq y$
  - (c)  $x > y$
  - (d)  $x \geq y$
  - (e) If either  $x=y$  or the relationship can't be established

**Answer (a)  $x < y$**

87. I.  $3x^2 + 12x - 180 = 0$   
II.  $2y^2 + 4y - 96 = 0$
- (a)  $x < y$
  - (b)  $x \leq y$
  - (c) If either  $x=y$  or the relationship can't be established
  - (d)  $x > y$
  - (e)  $x \geq y$

**Answer (a)  $x < y$**

**DIRECTIONS (Q. 88)**

In each of these questions two equations numbered I and II are given. You have to solve both the equations and give answer.

88. I.  $36x^2 + 30x + 6 = 0$  II.  $45y^2 + 24y + 3 = 0$
- (a)  $x < y$
  - (b)  $x \leq y$
  - (c) If either  $x=y$  or the relationship can't be established
  - (d)  $x > y$
  - (e)  $x \geq y$

**Answer (a)  $x < y$**

89. The series is  $2 \times 8 + 10 = 26$   $26 \times 6 - 12 = 144$   $144 \times 4 + 14 = 590$   $590 \times 2 - 16 = 1164$   $1164 \times 1 + 18 = 1182$  Directions (Questions 201 to 210): What should come in place of question mark (?) in the following number series? 201). 1 2 6 15 31 56 ?

- (a) 96
- (b) 94
- (c) 98
- (d) 92
- (e) 99 202). 41 39 43 37 45 ?

**Answer (e) 99 202). 41 39 43 37 45 ?**

**DIRECTIONS (Q. 90)**

What will come in place of question mark (?) in the given number series ?

90. 44 ? 99 148.5 222.75 334.125

- (a) 72
- (b) 77
- (c) 66
- (d) 54
- (e) 84 485). 7 8 4 13 -3 22 ?

**Answer (a) 72**

**DIRECTIONS (Q. 91)**

What approximate value should come in place of question mark (?) in the following questions? 321).

91.  $? \approx (24.02)^2 - 299 \div 12 \times 13.95 \approx 576 - 25 \times 14 = 576 - 350 = 226 \approx 225$  Directions (Q. 331-340): What should come in the place of question mark (?) in the following questions? 331).  $193.261 + 275.373 + 136.93 + 17.229 = ?$

- (a) 723.793
- (b) 622.793
- (c) 632.673
- (d) 593.603
- (e) 713.683 332).  $[(43)^{37} \times (43)^{-41}] / (43)^{-6} = ?$  333).  $(53.7)^2 - (43.6)^2 = ?$

**Answer (b) 622.793**

92.  $] = [(842 / 25) \times (940 / 27)] + [(616 / 52) \times (2089 / 19)] = 1172.562 + 1302.453 = 1165.14 + 1302.45 = 2475.01$  Directions (Q. 391-400): What should come in place of question mark in the following questions? 391).  $69\% \text{ of } 545 \div 2 = ?$

- (a) 178.082
- (b) 188.025
- (c) 198.075
- (d) 178.825
- (e) 168.025 392).  $(9/6) \text{ of } 642 + 97\% \text{ of } 846 + (5/7) \div (9/63) = ?$  393).  $(17)^2 - (18.2)^2 + (15)^2 = ?$

**Answer (e) 168.025 392). (9/6) of 642 + 97% of 846 + (5/7) ÷ (9/63) = ? 393). (17)² – (18.2)² + (15)²**

93. If the Yamaha bike's time taken is 4 hr, how much time taken by Honda bike?

- (a) 10 hr
- (b) 7.5 hr
- (c) 8 hr
- (d) 6.5 hr
- (e) None of these

**Answer (d) 6.5 hr**



94. The total time taken by Yamaha bike is 2 hr/km, how much time taken by TVS bike and Hero bike?
- 3.235 hr
  - 4.035 hr
  - 5.145 hr
  - 7.375 hr
  - None of these
- Answer (b) 4.035 hr**
95. If TVS bike distance is 78km then what is the difference between the time taken by Bajaj bike and Hero bike?
- 4.25 hr
  - 3.50 hr
  - 4.75 hr
  - 3.75 hr
  - None of these
- Answer (d) 3.75 hr**
96. The total investment of Dinesh, Mahesh and Suresh in 2008 is Rs 16,800. If difference in shares of profit of Mahesh and Dinesh is Rs 5250, find the investment of Mahesh.
- 6800
  - 7200
  - 5400
  - 3600
  - 8400
- Answer (e) 8400**
97. In 2009, if out of the total profit, Suresh got Rs 9750 as his share, what is the number of months for which Suresh invested his money?
- 4 months
  - 9 months
  - 6 months
  - 2 months
  - 3 months
- Answer (c) 6 months**
98. In 2007, Dinesh and Mahesh invested Rs 3900 and Rs 2600 respectively. Also Dinesh and Mahesh invested for same number of months. If difference in the shares of Dinesh and Mahesh out of total profit is Rs 2750, find the ratio of number of months of investment of Dinesh to investment of Suresh.
- 3 : 650
  - 2 : 270
  - 1 : 1040
  - 4 : 170
  - 1 : 740
- Answer (d) 4 : 170**
99. In 2010, if out of the total profit, Dinesh and Suresh got a share of Rs 10,400 and Rs 19,500 respectively, what is the ratio of investment of Dinesh, Mahesh and Suresh respectively?
- 3 : 4 : 5
  - 4 : 1 : 5
  - 4 : 3 : 5
  - 2 : 3 : 5
  - 1 : 5 : 3
- Answer (a) 3 : 4 : 5**

**100.** Upstream speed of Orange boat is 9 km/hr. How many more hours will it take to cover a distance of 315 km upstream than same distance downstream?

- (a) 18 hours
- (b) 25 hours
- (c) 22 hours
- (d) 20 hours
- (e) 15 hours

**Answer (c) 22 hours**

**101.** On a particular day, the time taken by Red boat to cover 255 km downstream is 2 hours more than the time taken by Yellow boat to cover 195 km upstream. Find the ratio of upstream speed to downstream speed in case of Red boat on that particular day.

- (a) 4 : 11
- (b) 3 : 5
- (c) 15 : 22
- (d) 8 : 19
- (e) 7 : 17

**Answer (e) 7 : 17**

**102.** Blue boat covered 756 km downstream on Wednesday for which it took 6 hours less than that in which it covered half distance upstream on Thursday. On Thursday, speed of stream was 2 km/hr more than that on Wednesday. Find the downstream speed of Blue boat on Thursday.

- (a) 21 km/hr
- (b) 19 km/hr
- (c) 23 km/hr
- (d) 11 km/hr
- (e) 15 km/hr

**Answer (a) 21 km/hr**

**103.** If D and F started to work and after 2 days they were replaced by Q, T and S then they stopped, how much work is still left? (Q, T and S worked for 2 days together)

- (a)  $\frac{1}{8}$
- (b)  $\frac{1}{6}$
- (c)  $\frac{1}{4}$
- (d)  $\frac{1}{10}$
- (e)  $\frac{1}{5}$

**Answer (d)  $\frac{1}{10}$**

**104.** E and C started work. E worked at 120% of his efficiency and C at  $\frac{5}{6}$  of his normal routine work. They started work on alternate days starting with E. How many days will be taken by them to complete the work?

- (a)  $\frac{65}{6}$  days
- (b)  $\frac{73}{6}$  days
- (c)  $\frac{80}{6}$  days
- (d)  $\frac{58}{6}$  days
- (e)  $\frac{54}{7}$  days

**Answer (a)  $\frac{65}{6}$  days**

**105.** Which of the following minister selected from Madhya Pradesh?

- (a) Harsh Vardhan
- (b) Rajnath Singh
- (c) Ananth Kumar
- (d) Suresh Prabhu
- (e) Jagat Prakash Nadda

**Answer (e) Jagat Prakash Nadda**

**106.** Which of the following statement is false?

- (a) Akram working in Palestine
- (b) David faces the person working in Iraq
- (c) The person working in Kuwait faces Vijay
- (d) There are two persons between vacant seat and Wahid
- (e) All the options are true

**Answer (c) The person working in Kuwait faces Vijay**

**107.** Which of the following model bike Karthik have?

- (a) TVS Akula
- (b) Suzuki Gixxer
- (c) Honda CB Hornet 160R
- (d) Royal Enfield Classic
- (e) KTM 1050 Adventure

**Answer (d) Royal Enfield Classic**

**108.** Who drive for Renault Sport team?

- (a) Max Verstappen
- (b) Fernando Alonso
- (c) Nico Rosberg
- (d) Lewis Hamilton
- (e) Sebastian Vettel

**Answer (a) Max Verstappen**

**109.** Which of the following statements is correct?

- (a) Valtteri Bottas won Australian Grand Prix Bank
- (b) Nico Rosberg drive for Scuderia Toro Rosso team
- (c) Fernando Alonso was born in the month of August
- (d) Lewis Hamilton was born in Belgium
- (e) None of these

**Answer (e) None of these**

**110.** Which of the following combinations is true? a) November – E – Male - 500

- (b) April – G – Female - 1600
- (c) August – B – Male - 2500
- (d) July – D - Female - 80
- (e) January – A - Female - 5000
- (a) 500

**Answer (d) July – D - Female - 80**

**111.** Punjab National Bank located in \_\_\_\_\_

- (a) Rajasthan
- (b) Gujarat
- (c) Maharashtra
- (d) Bihar
- (e) Karnataka

**Answer (a) Rajasthan**

**112.** Which of the following combinations is true? a) 3 – Syndicate Bank – Jharkhand - 500

- (b) 1 – Punjab National Bank – Karnataka - 700
- (c) 2 - Andhra Bank – Maharashtra - 1000
- (d) 7 – Canara Bank – Gujarat - 700
- (e) 4 – Vijaya Bank – Bihar - 900
- (a) Chandran and Vignesh

**Answer (e) 4 – Vijaya Bank – Bihar - 900**

113. Gandhi buys magazine for how much rupees?

- (a) 63
- (b) 61
- (c) 76
- (d) 96
- (e) 55

**Answer (a) 63**

114. The one who likes Fossil watch likes which of the following mobile?

- (a) Sony
- (b) Vivo
- (c) Apple
- (d) Oppo
- (e) Samsung

**Answer (c) Apple**

115. Which of the following is correctly matched?

- (a) N – Punjab –
- (b) Q – Odisha –
- (c) P – Maharashtra - SSC CHSL
- (d) O – Karnataka – SSC CHSL
- (e) None of these

**Answer (c) P – Maharashtra - SSC CHSL**

**DIRECTIONS (Q. 116)**

Study the following information carefully and answer the given questions. A number arrangement machine arranges two digit numbers into a typical manner. Each step gives output taking input from the previous step. The following is an illustration of Input and rearrangement. Using the illustration answer the question given below.  
136 Input: 3 5 8 3 4 6 4 5 2 1 2 4 Step I 8 9 9 5 8 8 Step II 5 0 4 4 Step III 1 -4 Step IV -3 Input:

116. Which of the following combination represent the first digit of the third number and second digit of the second number in step I of the given input?

- (a) 9, 9
- (b) 9, 8
- (c) 9, 4
- (d) 7, 9
- (e) 6, 5

**Answer (a) 9, 9**

**DIRECTIONS (Q. 117-118)**

Study the following information carefully and answer the given questions. A number arrangement machine arranges two digit numbers into a typical manner. Each step gives output taking input from the previous step. The following is an illustration of Input and rearrangement. Using the illustration answer the question given below.  
Input: 3 5 7 3 1 2 2 1 3 1 9 4 Step I 6 5 7 9 8 9 Step II 2 3 2 1 Step III 13 5 Step IV 18 Input: 2 3 7 3 1 2 9 5 2 1 1 1

117. Which of the following value to add with final input to make the value perfect square?

- (a) 17
- (b) 18
- (c) 13
- (d) 12
- (e) 11

**Answer (c) 13**

**118.** If both the numbers in the step II is multiply with 2 and then added, then what will be the final sum?

- (a) 98
- (b) 72
- (c) 64
- (d) 68
- (e) 48

**Answer (d) 68**

**DIRECTIONS (Q. 119)**

Study the following information carefully and answer the given questions. A number arrangement machine arranges two digit numbers into a typical manner. Each step gives output taking input from the previous step. The following is an illustration of Input and rearrangement. Using the illustration answer the question given below. 147 Input: 5 6 2 3 6 2 1 3 1 7 2 4 Step I 1 5 6 1 4 3 2 4 4 Step II 1 3 1 3 0 4 Step III 1 0 2 6 Step IV 1 6 Step V 7 Input:

**119.** If each digit in the step I is added with 2, then what will be the second number in same step?

- (a) 856
- (b) 748
- (c) 368
- (d) 582
- (e) 947

**Answer (e) 947**

**DIRECTIONS (Q. 120)**

Study the following information carefully and answer the given questions. A number arrangement machine arranges two digit numbers into a typical manner. Each step gives output taking input from the previous step. The following is an illustration of Input and rearrangement. Using the illustration answer the question given below. Input: 2 3 9 4 2 9 7 6 4 7 3 8 Step I 1 1 5 1 6 8 1 5 9 Step II 0 5 4 8 4 5 Step III 0 8 1 8 Step IV 1 0 Step V 1 Input: 7 5 8 3 2 6 9 5 2 4 1 8

**120.** If each digit in the step II is added with 3, then what will be the final sum of all digits in step II?

- (a) 28
- (b) 30
- (c) 36
- (d) 22
- (e) 18

**Answer (a) 28**

**121.** What is the possible word for “56&U, 12\$T” in the given code language?

- (a) for cousins
- (b) named pre
- (c) book cousins
- (d) societies viewed
- (e) None of these

**Answer (a) for cousins**

**122.** What is the code for ‘difficult’ in the given code language?

- (a) 03\*D
- (b) 13%N
- (c) 04\$E
- (d) 02@C
- (e) None of these

**Answer (c) 04\$E**

123. What is the possible word for “11%E,23%K” in the given code language?

- (a) abbreviations readjusts
- (b) backbenchers objectively
- (c) Pages objectively
- (d) That Pages
- (e) None of these

**Answer (c) Pages objectively**

124. What may be the possible word for ‘22\*B, 23#A,23!A” in the given code language?

- (a) coach huge layer
- (b) house layer people
- (c) cartoon hold succeeded
- (d) house hold succeeded
- (e) None of these

**Answer (d) house hold succeeded**

125. What may be the possible word for “02#Y,06%U,08\$S” in the given code language?

- (a) supermarket arranging worked
- (b) sales planned started
- (c) devika in reliance
- (d) reliance in mountains
- (e) None of these

**Answer (c) devika in reliance**

126. What is the code for ‘study’ in the given code language?

- (a) 20#R
- (b) 12#G
- (c) 06@X
- (d) 12#L
- (e) None of these

**Answer (a) 20#R**

127. What may be the possible code for ‘upto remember’ in the given code language?

- (a) 07#D, 11\$Q
- (b) 07#X, 15@U
- (c) 13&H,21!X
- (d) 09\*I, 09\*K
- (e) None of these

**Answer (b) 07#X, 15@U**

128. What may be the possible word for “49%H, 16#O,16#T ” in the given code language?

- (a) United nations provide
- (b) Government fund service
- (c) Government nations service
- (d) Union nations fund
- (e) None of these

**Answer (c) Government nations service**

129.  $(?)^2 = (2/7)$  of  $3693.83 \div 2.061$  e) None of these  $\approx (2/7) \times 3694 \times (1/2) = 528 \therefore ? = \sqrt{528} \approx \sqrt{23 \times 23} = 23$  283).  $52 \times 26 \div \sqrt{169} - 32 = ? - 42$

- (a) 65
- (d) 6
- (b) 57
- (e) None of these
- (c) 55 284).  $\sqrt{(54 \times 5 + 150 + 21)} = (?)^2$  c) 158 285).  $72 \times 492 = 7? \times 343$  b) 80.701

**Answer (c) 55 284).**  $\sqrt{(54 \times 5 + 150 + 21)} = (?)^2$  c) 158 285).  $72 \times 492 = 7? \times 343$  b) 80.701

**130.** ] 395).  $3\sqrt{(357911)} = ? = [ (842 / 25) \times (940 / 27) ] + [ (616 / 52) \times (2089 / 19)$  a) 71 ] b) 91 = 1172.562+1302.453 c) 31 = 1165.14 + 1302.45 = 2475.01 d) 81 Answer: e) e) 61

(b) 56200 391). 69% of  $545 \div 2 = ?$  c) 55200

(a) 178.082

(d) 58200

(e) 57240

(c) 198.075 392).  $(9/6)$  of 642 + 97% of 846 +  $(5/7) \div (9/63) = ?$  c) 56 393).  $(17)^2 - (18.2)^2 + (15)^2 = ?$  c) 4

**Answer (c) 198.075 392).**  $(9/6)$  of 642 + 97% of 846 +  $(5/7) \div (9/63) = ?$  c) 56 393).  $(17)^2 - (18.2)^2 +$

**131.** If the Honda bike distance is 400km, what is the total time taken by TVS bike, Hero bike, Bajaj bike, Suzuki bike?

(a) 60 hour 30 min

(b) 71 hour 50 min

(c) 77 hour 40 min

(d) 54 hour 10 min

(e) None of these

**Answer (a) 60 hour 30 min**

**132.** The difference between the distance of Bajaj bike and Suzuki bike is 40km, How many more hour taken by Bajaj bike than Suzuki bike?

(a) 5 hour 26 minutes

(b) 7 hour 25 minutes

(c) 10 hour 50 minutes

(d) 9 hour 55 minutes

(e) None of these Solutions

**Answer (c) 10 hour 50 minutes**

**133.** In 2006, out of the total profit, Mahesh got Rs 12,000 as his share. Suresh invested Rs 3600. If ratio of share of Suresh in total profit to total profit is 3 : 10, find the total profit.

(a) 30000

(b) 25000

(c) 35000

(d) 60000

(e) 40000 Solutions

**Answer (b) 25000**

**134.** For Green boat, its upstream speed is 6 km/hr on a particular day. Find the difference in time in covering 360 km by Red boat and Green boat on that particular day.

(a) 8 hours

(b) 6 hours

(c) 4 hours

(d) 7 hours

(e) 5 hours Solutions

**Answer (c) 4 hours**

**135.** If all the boys work together then time taken by them will be how much less than the time taken by P, R, Q and S together to do the work ?

(a) 9/5 days

(b) 13/5 days

(c) 11/5 days

(d) 14/5 days

(e) 17/5 days

**Answer (d) 14/5 days**

136. How many days will it take to complete the work if A does the 25% of the work alone and 10% is done by C alone and the remaining is done by U?

- (a) 21 days
- (b) 18 days
- (c) 19 days
- (d) 17 days
- (e) 25 days Solutions

**Answer (b) 18 days**

137. I.  $x^2 - 7\sqrt{3}x + 36 = 0$

II.  $y^2 - 11\sqrt{3}y + 84 = 0$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (d) if  $x < y$**

138. I.  $x^2 - 8x - 20 = 0$

II.  $3y^2 - 60y + 297 = 0$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (d) if  $x < y$**

139. I.  $2x^2 + 15x + 28 = 0$

II.  $2y^2 + 13y + 21 = 0$

- (a) if  $x > y$
- (b) if  $x \geq y$
- (c) if  $x < y$
- (d) if  $x \leq y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (c) if  $x < y$**

140. I.  $x^2 + (343)^{1/3} = 56$

II.  $(y)^{4/3} \times (y)^{5/3} - 295 = 217$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (d) if  $x < y$**

141. I.  $3x^2 - 10x + 8 = 0$

II.  $2y^2 - 19y + 35 = 0$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (d) if  $x < y$**



142. I.  $x^2 - 25x + 114 = 0$

II.  $y^2 - 10y + 24 = 0$

- (a) if  $x > y$
- (b) if  $x < y$
- (c) if  $x \geq y$
- (d) if  $x \leq y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (a) if  $x > y$**

143. I.  $25x^2 - 90x + 72 = 0$

II.  $y^2 + 26y + 168 = 0$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (a) if  $x > y$**

144. I.  $3x^2 - 8x - 16 = 0$

II.  $3y^2 - 19y + 28 = 0$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (d) if  $x < y$**

145. I.  $6x^2 - 13x - 44 = 0$

II.  $4y^2 - 17y - 42 = 0$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (d) if  $x < y$**

146. I.  $35x^2 + 4x - 63 = 0$

II.  $7y^2 - 4y - 20 = 0$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (d) if  $x < y$**

147. I.  $x^2 - 4\sqrt{7}x + 21 = 0$

II.  $2y^2 - 8\sqrt{5}y - 50 = 0$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (a) if  $x > y$**

148. I.  $3x^2 - 5x - 12 = 0$

II.  $2y^2 + 15y + 25 = 0$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (a) if  $x > y$**

149. I.  $x^2 - 4(\sqrt{2} + \sqrt{5})x + 16\sqrt{10} = 0$

II.  $y^2 - 5(\sqrt{3} + 2\sqrt{2})y + 50\sqrt{6} = 0$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (d) if  $x < y$**

150. I.  $5x^2 + 11x - 12 = 0$

II.  $4y^2 - 13y - 12 = 0$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (a) if  $x > y$**

151. I.  $2x^2 + 3x - 20 = 0$

II.  $2y^2 + 15y + 28 = 0$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (a) if  $x > y$**

152. I.  $x^2 - 13.5x + 38 = 0$

II.  $y^2 - 1.5y - 10 = 0$

- (a) if  $x > y$
- (b) if  $x \leq y$
- (c) if  $x \geq y$
- (d) if  $x < y$
- (e) if  $x = y$  or relationship between  $x$  and  $y$  can't be established

**Answer (d) if  $x < y$**

**DIRECTIONS (Q. 153)**

With an increased variability of monsoons and rapidly depleting groundwater tables, large parts of India are reeling under water stress and drought like situation. Here, one intervention that has been tried out in various States, and perhaps needs to be taken up on a bigger scale, is the construction of farm ponds, it can be cost-effective structures that transform rural livelihoods. They can help enhance water control, contribute to agriculture intensification and boost farm incomes.

**153.** Which of the following statement(s) is/are false in context of the given statement?

- (a) The problem of lower water table has captivated the entire country
- (b) Farm ponds can help in the process of water control
- (c) Construction of farm ponds hamper the agricultural modification & rural subsistence
- (d) Farm ponds can be a viable investment option
- (e) None of these

**Answer (c) Construction of farm ponds hamper the agricultural modification & rural subsistence**

**DIRECTIONS (Q. 154-155)**

What approximate value should come in place of question mark (?) in the following questions?

**154.**  $? = 729.11 \div 26.97 \times 81.11 \div 35.91 \times 108.18$

- (a) 2187
- (b) 6561
- (c) 729
- (d) 243
- (e) 81

**Answer (b) 6561**

**155.**  $47.87\% \text{ of } 350 + 60.11\% \text{ of } 280 = 96.98\% \text{ of } 299.78 + ?\% \text{ of } 150$

- (a) 30
- (b) 40
- (c) 50
- (d) 80
- (e) 45

**Answer (a) 30**

**156.** If in E INC got 15000 more valid votes than AAP and BJP won the election by 15000 votes, then find the valid votes received by BJP. (There are only three parties in E – BJP, INC & AAP)

- (a) 40000
- (b) 38000
- (c) 45000
- (d) 35000
- (e) 30000

**Answer (d) 35000**

**157.** The average wt. of boys in school is 60kg while average wt. of girls is 55 kg. The average wt. of both boys and girls is 58kg. Find the number girls in school if number of boys is 720.

- (a) 480
- (b) 720
- (c) 240
- (d) 360
- (e) 600

**Answer (a) 480**

**158.** If 21 is added in a number, the result becomes  $116\frac{2}{3}\%$  of itself. Find the new number?

- (a) 126
- (b) 147
- (c) 130
- (d) 136
- (e) 125

**Answer (b) 147**

**159.** An article when sold at  $\frac{4}{5}$  of its original selling price, gives a profit of 20%. Find the profit % when the same article is sold at its actual selling price.

- (a) 15 %
- (b) 20 %
- (c) 25%
- (d) 22 %
- (e) None of these

**Answer (e) None of these**

**160.** An amount of Rs. 20000 when invested at R% simple interest for 2 years becomes Rs. 24000. What will it become in 3 years if invested at  $(R+2)\%$ ? (in Rs.)

- (a) 27200
- (b) 26300
- (c) 25200
- (d) 27400
- (e) 28100

**Answer (a) 27200**

**161.** Karan purchased an article marked up by 50% at a discount of 20% but later he found that the article was having a defect so he decided to return it but the shopkeeper returned him only 90% of what he had paid. What is profit/loss (in %) of the shopkeeper in the whole transaction?

- (a) 8
- (b) 10
- (c) 12
- (d) 15
- (e) None of these

**Answer (c) 12**

**162.** A sum of Rs. x was invested at 10% simple interest for 3 years. If the same sum was invested at 4% more for same period, then it would have fetched Rs. 120 more. Find the value of 5x. (in Rs.)

- (a) 5000
- (b) 4800
- (c) 3600
- (d) 5500
- (e) 4000

**Answer (a) 5000**

**163.** A sum of Rs. P was invested at 10% for 2 years at simple interest. If the same sum was invested at 20% for 'x' years, it would have fetched Rs. 200 more. Find 'x' if  $Px = 5000$ . (value of x is given in months)

- (a) 12
- (b) 18
- (c) 15
- (d) Cannot be determined
- (e) None of these

**Answer (c) 15**

**164.** A sum of Rs 1400 becomes Rs 2408 in 8 yrs at simple interest, then find the rate of interest for last 4 yrs, if the interest rate for 1st 4 yrs is 12% per annum ?

- (a) 8 %
- (b) 10 %
- (c) 6%
- (d) 4 %
- (e) None of these

**Answer (c) 6%**

**165.** If the shopkeeper marked the price of an item 60% above the cost price and then gives two successive discount of 10% and 15% respectively, then find the profit percentage of the shopkeeper on selling the item?

- (a) 25.4%
- (b) 22.4%
- (c) 20%
- (d) 28.5%
- (e) 32%

**Answer (b) 22.4%**

**166.** Johnny calculates his profit at cost price while Jini at selling price. If cost price is same for all and everyone calculate their profit as 10%. Find ratio of selling price.

- (a) 100:111
- (b) 10:11
- (c) 10:101
- (d) 99:100
- (e) Cannot be determined

**Answer (d) 99:100**

**167.** An amount doubles in 5 years at simple interest. In what time will it become 12 times of itself at same rate? (in years)

- (a) 30
- (b) 50
- (c) 55
- (d) 36
- (e) None of these

**Answer (c) 55**

**168.** An amount of 4000 rs is invested at 20% per annum for 2 yrs at compound interest compounding half-yearly, then find the total interest amount received after 2 yrs ?

- (a) Rs 1856.4
- (b) Rs 1812.4
- (c) Rs 1882.4
- (d) Rs 1912.4
- (e) None of these

**Answer (a) Rs 1856.4**

**169.** Anu bought a purse at a discount of 20% which was marked at 30% higher than cost price. A customer gets 10% extra discount on purchase of more than two purses. If Anu purchased 3 such purses, find profit or loss percent of shopkeeper.

- (a) 4% loss
- (b) 4% profit
- (c) 6.4% profit
- (d) 6.4% loss
- (e) None of these

**Answer (d) 6.4% loss**

**170.** A train crosses a tunnel which is half of its length with a speed of 144 km/hr. in  $\frac{1}{2}$  min, then find the time in which it will cross another train which is double of its length and standing on platform in opposite direction with 60% of its initial speed ?

- (a) 120 sec
- (b) 90 sec
- (c) 150 sec
- (d) 100 sec
- (e) 180 sec

**Answer (d) 100 sec**

**171.** How many students are there in Section B of class?

- (a) 50
- (b) 60
- (c) 90
- (d) 100
- (e) 110

**Answer (a) 50**

**172.** what is average of students who passed in only one examination in all sections together?

- (a) 39.67
- (b) 40.67
- (c) 41.67
- (d) 42.67
- (e) 43.67

**Answer (c) 41.67**

**173.** What is the average no. of magazines distributed by companies Q, R and T among their respective distributors?

- (a) 2720
- (b) 2640
- (c) 2480
- (d) 2960
- (e) 3120

**Answer (c) 2480**

**174.** Find the respective ratio of the total number of magazines distributed among the distributors of company R to that of the total no. of magazines distributed among distributors of company T?

- (a) 21:19
- (b) 19:21
- (c) 17:21
- (d) 21:17
- (e) 17:23

**Answer (b) 19:21**

**175.** Find the average number of books distributed among the distributors by all the five companies together?

- (a) 2784
- (b) 2664
- (c) 2680
- (d) 2756
- (e) 2724

**Answer (e) 2724**

**DIRECTIONS (Q. 176-178)**

Study the paragraph carefully and answer the following questions. A publishing house sold three books i.e. Quant, English and reasoning on three different stores i.e. A, B and C. Quant, reasoning and English book are sold at 20% discount by store A, C and B respectively. Quant, reasoning and English book are sold at 15% discount by store C, B and A respectively. Discount percent given on Quant book by store B is half of discount percent given on reasoning book by store C. M.R.P. for each book is same at every store.

**176.** If market price of a reasoning book was 50% more than cost price of the book for store C. Find profit percent on selling a reasoning book by store C?

- (a) 20%
- (b) 15%
- (c) 25%
- (d) 10%
- (e) 12.5%

**Answer (a) 20%**

**177.** What is the ratio of average discount given on quant book by store A, reasoning book by store C and English book by store B to market price of a book?

- (a) 5:4
- (b) 2:3
- (c) 3:2
- (d) 1:5
- (e) 4:5

**Answer (d) 1:5**

**178.** If an English book is sold at Rs. 170 by store A, the find selling price of reasoning book for store C?

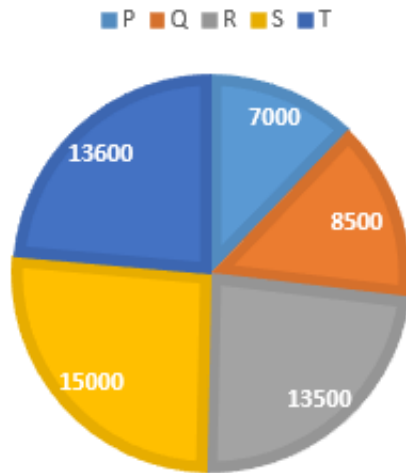
- (a) Rs. 160
- (b) Rs. 170
- (c) Rs. 135
- (d) Rs. 105
- (e) None of these

**Answer (a) Rs. 160**

**DIRECTIONS (Q. 179-182)**

Pie chart given below shows no. of non-defective article manufactured by five different firms i.e. P, Q, R, S and T.

**NO. OF NON-DEFECTIVE ARTICLE**



**179.** No. of non-defective article manufactured by firm Q is what percent of non-defective article manufactured by firm T?

- (a) 60%
- (b) 62.5%
- (c) 37.5%
- (d) 50%
- (e) 72.5%

**Answer (b) 62.5%**

**180.** If ratio of non-defective to defective article manufactured by firm S is 75:2, then find ratio of non-defective article manufactured by firm T to defective article manufactured by firm S?

- (a) 37:1
- (b) 34:1
- (c) 33:1
- (d) 24:1
- (e) 38:1

**Answer (b) 34:1**

**181.** Average no. of non-defective article manufactured by firm Q and R is what percent of total no. of non-defective article manufactured by firm S?

- (a)  $72\frac{2}{3}\%$
- (b)  $66\frac{2}{3}\%$
- (c)  $33\frac{1}{3}\%$
- (d)  $73\frac{1}{3}\%$
- (e) None of these

**Answer (d)  $73\frac{1}{3}\%$**

**182.** What is the ratio of non-defective article manufactured by firm Q and R together to that of P and S together?

- (a) 2:1
- (b) 1:1
- (c) 1:2
- (d) 1:3
- (e) 3:1

**Answer (b) 1:1**



**183.** 4 men & 3 children completes a project for Rs. 600 in 3 days. If a man completes same project in 15 days. Find daily wage of a man.

- (a) Rs 36
- (b) Rs.40
- (c) Rs.44
- (d) Rs.48
- (e) Rs.42

**Answer (b) Rs.40**

**184.** Difference between 50% of y and 10% of x is 170 whereas difference between 40% of x and 30% of y is zero. Find the sum of 'x' and 'y' ?

- (a) 770
- (b) 630
- (c) 600
- (d) 700
- (e) 560

**Answer (d) 700**

**185.** If ratio of time periods of investment of P and Q is 4:5, profit at the end of the year is 75000 and P's share is Rs 15000, then what is the ratio of Q's and P's investment?

- (a) 5:16
- (b) 6:7
- (c) 12:13
- (d) 16:5
- (e) 8:5

**Answer (d) 16:5**

**186.** Average of 8 consecutive odd numbers is 10. What will be the average of smallest 4 numbers out of 8 numbers?

- (a) 7
- (b) 8
- (c) 6
- (d) 4
- (e) 5

**Answer (c) 6**

**187.** The work done by 5 boys in 20 days can be done by 10 men in 8 days. 4 Men & 4 boys undertook a work to complete in 3 days for Rs. 540. Find the amount earned by boys for their whole contribution.

- (a) Rs 236
- (b) Rs.240
- (c) Rs.244
- (d) Rs.248
- (e) Rs.242

**Answer (b) Rs.240**

**188.** Sanjay scored 56% marks and passed an exam by 10 marks while Rohit scored 48% marks but failed by 6 marks. What is the pass percentage?

- (a) 52.5%
- (b) 51.5%
- (c) 52%
- (d) 51%
- (e) None of these

**Answer (d) 51%**

**189.** Four books are to be distributed among seven students. If no student gets more than one book, then the number of ways possible to do it is?

- (a) 180
- (b) 240
- (c) 260
- (d) 210
- (e) 220

**Answer (d) 210**

**190.** In a basket, there are 8 red balls and 6 green balls. If 2 balls are taken out from the basket, then find what is the probability of both balls being either red or green?

- (a)  $\frac{43}{91}$
- (b)  $\frac{47}{91}$
- (c)  $\frac{51}{91}$
- (d)  $\frac{43}{87}$
- (e)  $\frac{43}{82}$

**Answer (a)  $\frac{43}{91}$**

**191.** The parallel sides of a trapezium are 4 cm & 10 cm respectively while non-parallel sides are equal to side of square of area 25 sq.cm. Find area of trapezium. (in sq.cm.)

- (a) 50
- (b) 42
- (c) 56
- (d) 28
- (e) 14

**Answer (d) 28**

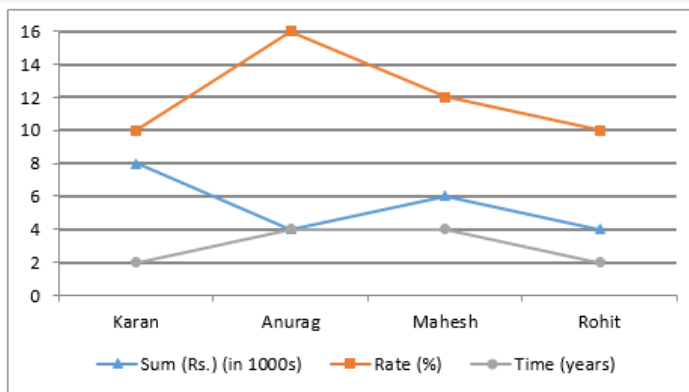
**192.** The ratio of area of square to that of rectangle of length 10 cm is 4 : 5. If breadth of rectangle is same as side of square. Find length of diagonal of square.

- (a)  $9\sqrt{2}$  cm
- (b)  $10\sqrt{2}$  cm
- (c)  $6\sqrt{2}$  cm
- (d)  $4\sqrt{2}$  cm
- (e)  $8\sqrt{2}$  cm

**Answer (e)  $8\sqrt{2}$  cm**

**DIRECTIONS (Q. 193-197)**

Given line graph shows the sum invested, rate of interest and time period of investment by 4 people. Study the data carefully and answer the questions. (NOTE: all invested their sum at simple interest)



**193.** How much will Rohit receive after completion of his investment period? (in Rs.)

- (a) 5200
- (b) 6800
- (c) 4800
- (d) 4400
- (e) 4600

**Answer (c) 4800**

**194.** Interest amount received by Mahesh is what percent more than interest amount received by Karan?

- (a) 85%
- (b) 60%
- (c) 75%
- (d) 70%
- (e) 80%

**Answer (e) 80%**

**195.** What is total amount received as interest by Anurag & Rohit together? (in Rs.)

- (a) None of these
- (b) 3150
- (c) 3200
- (d) 3360
- (e) 3420

**Answer (d) 3360**

**196.** If Karan had invested same sum at compound interest at same rate of interest for same period. How much more would he earn?

- (a) Rs 80
- (b) Rs 90
- (c) Rs 70
- (d) Rs 60
- (e) None of these

**Answer (a) Rs 80**

197. Who among the four had received the highest amount as interest?

- (a) Karan
- (b) Anurag
- (c) Both Anurag & Mahesh
- (d) Rohit
- (e) Mahesh

**Answer (e) Mahesh**

**DIRECTIONS (Q. 198-202)**

Following Table chart gives the details of 5 students of a particular school in five different subjects in the annual exam. Note:-the data provided in the table is percentage of marks out of total marks in that particular subject.

|           | Maths(150) | Physics(150) | Chemistry(150) | English(100) | Computer(100) |
|-----------|------------|--------------|----------------|--------------|---------------|
| Amit      | 70         | 66           | 58             | 54           | 80            |
| Aakash    | 50         | 64           | 78             | 65           | 75            |
| Siddharth | 48         | 72           | 88             | 70           | 86            |
| Lokesh    | 80         | 76           | 84             | 75           | 85            |
| Ritesh    | 76         | 82           | 64             | 72           | 94            |

198. Total marks scored by Lokesh in physics, chemistry and maths together is how much more/less than total marks scored by Amit in the same three subjects together?

- (a) 75
- (b) 65
- (c) 69
- (d) 55
- (e) 80

**Answer (c) 69**

199. Find the overall percentage of marks scored by Siddharth in the exam?

- (a) 75%
- (b) 82%
- (c) 68%
- (d) 72%
- (e) 80%

**Answer (d) 72%**

200. Find the difference of total marks scored by Ritesh in all the given subject together and total marks scored by Aakash in all the given subjects together?

- (a) 71
- (b) 84
- (c) 78
- (d) 82
- (e) 93

**Answer (a) 71**

201. Find the average marks scored in physics subject by all the given five students together?

- (a) 105
- (b) 110
- (c) 108
- (d) 100
- (e) 98

**Answer (c) 108**

**202.** Total marks scored by Aakash, Siddharth and Lokesh in English is what percentage of the total marks scored by Amit, Aakash and lokesh in maths?

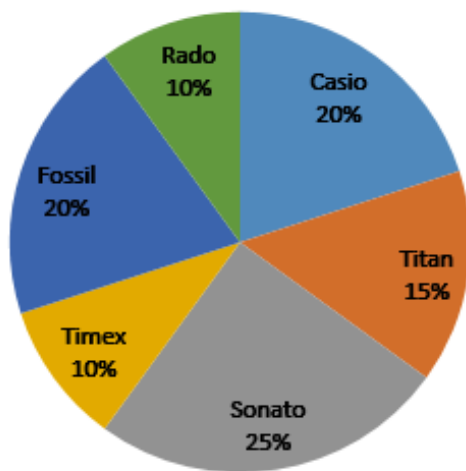
- (a) 75%
- (b) 70%
- (c) 65%
- (d) 68%
- (e) 80%

**Answer (b) 70%**

**DIRECTIONS (Q. 203-207)**

Given pie graph shows percentage distribution of watches manufactured by a company in 2018. Study the graph carefully & answer the questions.

**Total watches manufactured = 1000**



**203.** What is average of watches manufactured by Casio, Titan & Sonata together?

- (a) 500
- (b) 600
- (c) 400
- (d) 200
- (e) 300

**Answer (d) 200**

**204.** What is ratio of watches manufactured by Timex & Sonata together to that by Fossil & Casio together?

- (a) 5:4
- (b) 8:7
- (c) 7:8
- (d) 7:4
- (e) 5:8

**Answer (c) 7:8**

**205.** Watches manufactured of Sonata are what percent more/less than watches manufactured of Rado?

- (a) 130%
- (b) 150%
- (c) 200%
- (d) 170%
- (e) 100%

**Answer (b) 150%**

**206.** If next year, Titan watch production increases by 10% while that of Timex decreases by 10%. What is difference in manufacturing of both in next year?

- (a) 80
- (b) 90
- (c) 100
- (d) 65
- (e) 75

**Answer (e) 75**

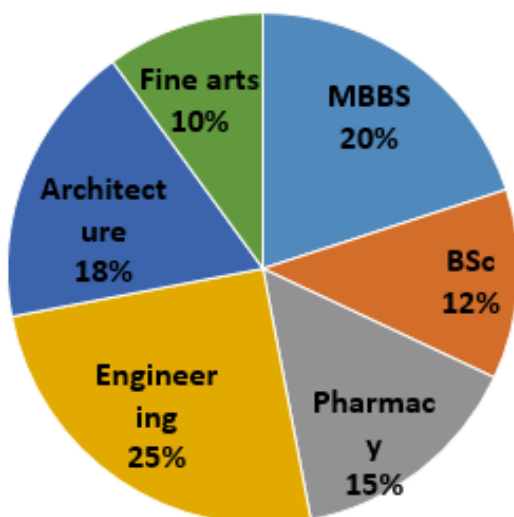
**207.** No. of watches manufactured of how many brands is more than average no. of watches manufactured?

- (a) 4
- (b) 3
- (c) 1
- (d) 2
- (e) 5

**Answer (b) 3**

**DIRECTIONS (Q. 208-212)**

Study the charts given below carefully and answer the following questions. Pie chart shows the distribution of total students of a university in different departments as shown below.



**208.** Find the ratio of total number of student from engineering and architecture department together to total students from the pharmacy and BSc department together?

- (a) 27:43
- (b) 27:47
- (c) 43:27
- (d) 47:27
- (e) 37:42

**Answer (c) 43:27**

**209.** If the ratio of males to females in pharmacy and finearts departments are 1:2 and 3:2 respectively, then find the total number of females in pharmacy and finearts together?

- (a) 784
- (b) 712
- (c) 736
- (d) 756
- (e) 812

**Answer (d) 756**

**210.** Find the central angle of the total students of architecture department of the university?

- (a)  $64.8^\circ$
- (b)  $75.6^\circ$
- (c)  $72^\circ$
- (d)  $43.2^\circ$
- (e)  $68.4^\circ$

**Answer (a)  $64.8^\circ$**

**211.** Number of students who failed in the final semester exam from MBBS and Finearts dept are 20% and 15% respectively of their respective dept, then find the total number of student who passed the semester from MBBS and Finearts dept ?

- (a) 1345
- (b) 1323
- (c) 1368
- (d) 1420
- (e) 1456

**Answer (b) 1323**

**212.** Total students from engineering and pharmacy department together is approximately what percentage of the total students from MBBS and Fine arts dept?

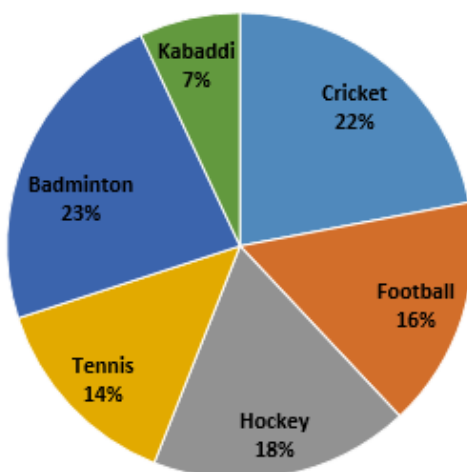
- (a) 122%
- (b) 148%
- (c) 126%
- (d) 143%
- (e) 134%

**Answer (e) 134%**

**DIRECTIONS (Q. 213-217)**

Study the charts given below carefully and answer the following questions. Pie chart shows the percentage distribution of total Spectators of a particular city loving different sports as shown below.

**Total Spectators = 14000**



**213.** Total spectators of Badminton and kabaddi together is what percentage of total spectators of cricket and hockey together ?

- (a) 70%
- (b) 75%
- (c) 80%
- (d) 65%
- (e) 60%

**Answer (b) 75%**

**214.** Find the ratio of total spectators of Football and tennis together to the total spectators of Cricket?

- (a) 17:12
- (b) 11:15
- (c) 15:11
- (d) 12:17
- (e) 13:18

**Answer (c) 15:11**

**215.** Find the central angle of total spectators of badminton and tennis together?

- (a)  $79.2^\circ$
- (b)  $136.8^\circ$
- (c)  $115.2^\circ$
- (d)  $126^\circ$
- (e)  $133.2^\circ$

**Answer (e)  $133.2^\circ$**

**216.** Out of total hockey spectators, male and female lovers are in the ratio 9: 6 respectively, then find difference between male and female spectators of hockey?

- (a) 524
- (b) 484
- (c) 336
- (d) 504
- (e) 472

**Answer (d) 504**

**217.** Total spectators of cricket and football together is how much more/less than total spectators of badminton and tennis together?

- (a) 160
- (b) 140
- (c) 180
- (d) 200
- (e) None of these

**Answer (b) 140**

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**218.** If in D there are only two parties-BJP & INC and BJP got 70% of the total valid votes, then find by how many votes BJP won the election in D.

- (a) 18000
- (b) 25000
- (c) 10000
- (d) 16000
- (e) 6000 2

**Answer (a) 18000**



**219.** The interest earned on an amount after 2 yrs at 10 % per annum compounded yearly is Rs

- (a) Rs 1792
- (b) Rs 1864
- (c) Rs 1912
- (d) Rs 1754
- (e) Rs 1720

**Answer (a) Rs 1792**

**220.** Rs. 12000 becomes Rs. 15000 in 18 months at a certain rate of interest at simple interest. Find amount if Rs. 5000 invested at same rate for 30 months at simple interest.

- (a) Rs. 7883.33
- (b) Rs. 7083.33
- (c) Rs. 7279.80
- (d) Rs. 7173.33
- (e) None of these

**Answer (b) Rs. 7083.33**

**221.** Aakash sells an article at a profit of 10%. Had he bought it for 5% less and sold it for 120 rs more then he would have gained 20% profit. What is the cost price of the article ?

- (a) Rs 2500
- (b) Rs 4000
- (c) Rs 3000
- (d) Rs 3500
- (e) Rs 2000

**Answer (c) Rs 3000**

**222.** Arun sells his watch at a profit of  $33\frac{1}{3}\%$  & his purse at a loss of  $16\frac{2}{3}\%$  & on whole he gains Rs.

- (a) 53 : 15  $33\frac{1}{3}\%$  then there will be no profit no loss. Find cost price
- (b) 44 : 17
- (c) 41 : 17 of the watch ?
- (d) 47 : 15
- (e) 44 : 15

**Answer (d) 47 : 15**

**223.** Prabhat invested Rs. 15600 on SI at rate of R% p.a. for 3 years & the interest obtained is Rs. 7020. If he invested the same amount at rate of (R+5)% p.a. for two years on CI then find the interest obtained by Prabhat?

- (a) Rs. 6864
- (b) Rs. 6250
- (c) Rs. 6748
- (d) Rs. 6468
- (e) Rs. 6648

**Answer (a) Rs. 6864**

**224.** Students failed in both exams in section C are what percent of total students in section C? (in %)

- (a) 30
- (b) 20
- (c) 18
- (d) 25
- (e) 33.33 12

**Answer (e) 33.33 12**

**225.** Find the total numbers of distributors of magazines of company Q and T together?

- (a) 62
- (b) 72
- (c) 84
- (d) 78
- (e) 64 13 42, 142, 86, 91, 338, ? 84, ?

**Answer (d) 78**

**DIRECTIONS (Q. 226)**

Study the paragraph carefully and answer the following questions. A publishing house sold three books i.e. Quant, English and reasoning on three different stores i.e. A, B and C. Quant, reasoning and English book are sold at 20% discount by store A, C and B respectively. Quant, reasoning and English book are sold at 15% discount by store C, B and A respectively. Discount percent given on Quant book by store B is half of discount percent given on reasoning book by store C. M.R.P. for each book is same at every store.

**226.** If total selling price of Quant book for store A and B together is Rs.

- (a) Rs.240
- (b) Rs.270
- (c) Rs.280
- (d) Rs.300
- (e) Rs.600

**Answer (d) Rs.300**

**227.** Store A sold reasoning book at Rs. 880, find M.R.P of the book if discount given by store A on reasoning book is 20% more than discount given by store B on quant book?

- (a) Rs. 1200
- (b) Rs. 1000
- (c) Rs. 960
- (d) Rs. 1240
- (e) None of these

**Answer (b) Rs. 1000**

**228.** Average of 8 consecutive odd numbers is

- (a) 7
- (b) 8
- (c) 6
- (d) 4
- (e) 5

**Answer (c) 6**

**229.** The work done by 5 boys in 20 days can be done by 10 men in 8 days. 4 Men & 4 boys undertook a work to complete in 3 days for Rs.

- (a) Rs 236
- (b) Rs.240
- (c) Rs.244
- (d) Rs.248
- (e) Rs.242

**Answer (b) Rs.240**

**230.** Find the difference of total marks scored by Ritesh in all the given subject together and total marks scored by Aakash in all the given subjects together?

- (a) 71
- (b) 84
- (c) 78
- (d) 82
- (e) 93 18

**Answer (a) 71**

**231.** Find the central angle of total spectators of badminton and tennis together?

- (a)  $79.2^\circ$
- (b)  $136.8^\circ$
- (c)  $115.2^\circ$
- (d)  $126^\circ$
- (e)  $133.2^\circ$

**Answer (e)  $133.2^\circ$**

**DIRECTIONS (Q. 232)**

What will come in place of the question mark (?) in the following questions ?

**232.**  $[2.5 \div 0.1]3 \div (6.25)2 \times [(1.25)2 \div (5)2] = (25) 3+x$

- (a) 1
- (b)  $\frac{2}{3}$
- (c)  $\frac{1}{3}$
- (d) 2
- (e) None of these 1 161 1595 7 13. 377  $\times 253 \div 13 = (x) 3 6$

**Answer (c)  $\frac{1}{3}$**

**DIRECTIONS (Q. 233-239)**

What approximate value will come in place of (x or ?) in the following questions ? x 1.

**233.**  $(23.89)^2 + \sqrt{3598.97} - 171.93 + (?)^2 = (27.98)^2 + 4$

- (a) 18
- (b) 24
- (c) 28
- (d) 22
- (e) 26

**Answer (c) 28**

**234.**  $(13.97)^2 - 39.87\% \text{ of } 239.97 + (29.87)^2 + 330.97 = (?)^3$

- (a) 18
- (b) 19
- (c) 17
- (d) 11

(e)  $15 ? + 134.5 12. + 209.87 + (69.87)^2 - 1999.83 = (54.87)^2 + 99.85 24 8$

**Answer (c) 17**

**235.**  $? \times 12.97 + (24.97)^2 - (19.89)^2 + 363.83 = (27.93)^2$

- (a) 10
- (b) 20
- (c) 18
- (d) 12

(e)  $15 593.89 1259.81 17.93 = (25.89)2 14. + 14.87 \times 35.88 + ?$

**Answer (d) 12**

**236.**  $48.01\% \text{ of } 249.99 + x\% \text{ of } 499.99 = (11.01)^2 + (2.11)^2$

- (a) 2
- (b) 3
- (c) 5
- (d) 1
- (e) 6 2

**Answer (c) 5**

**237.**  $49.11 \times 7 \div 14.09 + \sqrt{x} = 1025 \div 257$

- (a) 9
- (b) 8
- (c) 27
- (d) 16
- (e)  $25 \frac{3}{4} \times (15.99) + 31.9\% \text{ of } 3199 + 294.9 = x \times 17$

**Answer (a) 9**

**238.**  $? \% \text{ of } 6519.97 + (21.97)^2 - 29.98\% \text{ of } 9139.93 = (25.91)^2$

- (a) 25
- (b) 15
- (c) 30
- (d) 45
- (e)  $40 \frac{577.93}{?}$

**Answer (d) 45**

**239.**  $32.87\% \text{ of } ? - 17.87 \times 44.86 - \sqrt{399.81} = (24.98)^2 + 29.89$

- (a) 4450
- (b) 4500
- (c) 4350
- (d) 4600
- (e)  $4900 \frac{9}{?}$

**Answer (b) 4500**

**DIRECTIONS (Q. 240-244)**

Find the wrong number in the given series that does not follow the pattern? 1.

**240.** 325, 546, 754, 936, 1078, 1170

- (a) 936
- (b) 546
- (c) 325
- (d) 1078
- (e) 1170

**Answer (c) 325**

**241.** 192, 202, 210, 216, 225, 243, 288

- (a) 210
- (b) 202
- (c) 288
- (d) 192
- (e) 243

**Answer (c) 288**

**242.** 3, 11, 49, 191, 569, 1135, 1134

- (a) 1135
- (b) 1134
- (c) 3
- (d) 49
- (e) 11

**Answer (d) 49**

**243.** 23, 30, 42, 63, 95, 140, 200

- (a) 42
- (b) 140
- (c) 30
- (d) 200
- (e) 23

**Answer (a) 42**

**244.** 8, 30, 60, 120, 180, 180, 90

- (a) 30
- (b) 120
- (c) 90
- (d) 8
- (e) 60 11

**Answer (c) 90**

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**DIRECTIONS (Q. 245-246)**

What should come in place of the question mark (?) in the following series ? 1.

**245.** 141, 197, 269, 359, 469, ?

- (a) 607
- (b) 609
- (c) 601
- (d) 603
- (e) 605

**Answer (c) 601**

**246.** Two container P and Q have mixture. In P three liquid A, B and C are mixed in a ratio of 2 : 1 : 1 and in Q two liquid B and C are mixed in ratio of 3 : 2. Mixture of both container P and Q poured into third container X which show the quantity ratio of A, B and C is 8 : 13 : 10. Find the quantity in Q is what % less or more than the quantity in P.

- (a) 3.25%
- (b) 12.5%
- (c) 6.25%
- (d) 11%
- (e) 25%

**Answer (c) 6.25%**

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**DIRECTIONS (Q. 247-248)**

Three milkman P, Q and R have mixture of milk and water in the quantity of  $(X + 24)$  liters,  $(X+54)$  liters and  $(X + 84)$  liters and milk and water in the ratio of  $2 : 3$ ,  $8 : 7$  and  $13 : 5$  respectively.

- 247.** If P sold 50 liters of his mixture and 8 liters of milk added in remaining mixture, new ratio of milk and water becomes  $6 : 7$ . What quantity of water should be added by Q and R, so new ratio of their mixture becomes  $5 : 7$  &  $5 : 4$  respectively ?(1 marks)

- (a) 56 liters& 40 liters
- (b) 54 liters& 42 liters
- (c) 50 liters& 30 liters
- (d) 60 liters& 50 liters
- (e) 64 liters& 30 liters

**Answer (b) 54 liters& 42 liters**

- 248.** Q sold 40% of his mixture and R sold 50% of his mixture, remaining mixture of both became equal. If all three charge Rs. 2 for one liter of water and production cost of milk for P, Q and R in the ratio of  $5 : 6 : 7$  and total mixture, which all three milkman have cost Rs. 3644. Find production cost of milk per liter for P, Q and R respectively? (1 marks)

- (a) 5 Rs. 6 Rs & 7 Rs
- (b) 25 Rs. 30 Rs & 36 Rs
- (c) 10 Rs. 12 Rs & 16 Rs
- (d) 10 Rs. 12 Rs & 14 Rs
- (e) 10 Rs. 14 Rs& 15s

**Answer (d) 10 Rs. 12 Rs & 14 Rs**

- 249.** Vessel A & B contains mixture of orange juice and mango juice in the ratio of  $5 : 4$  and  $5 : 2$  respectively. Some quantity of mixture taken out from vessel A and B and mixed in another vessel C, if vessel C contains mixture of orange juice and mango juice in the ratio of  $5 : 3$  and total quantity of mixture in vessel C is 96 l. then find the difference between mixture taken out from vessel A & B?

- (a) 16 l
- (b) 12 l
- (c) 8 l
- (d) 10 l
- (e) 18 l

**Answer (b) 12 l**

- 250.** Mixture of petrol and water in vessel A are in the ratio of  $4:1$  and in vessel B are in the ratio of  $5:3$ . If 20 l of mixture taken out from vessel A and proud in vessel B so new ratio of petrol to water in vessel B becomes  $19:10$ , then find initially quantity of petrol in vessel A, if new quantity of mixture in vessel B is 26 l more than initial quantity of mixture in vessel A?

- (a) 64l
- (b) 72l
- (c) 68l
- (d) 56l
- (e) 48L

**Answer (b) 72l**

**251.** Two container P and Q have mixture of different liquids. In P three liquid A, B and C is mixed in a ratio of 2 : 1 : 1 and in Q two liquid B and C is mixed in ratio of 3 : 2. Mixture of both container P and Q poured into third container X which show the quantity ratio of A, B and C is 8 : 13 : 10. Find the quantity in Q is what % less or more than the quantity in P.

- (a) 3.25%
- (b) 12.5%
- (c) 6.25%
- (d) 11%
- (e) 25%

**Answer (c) 6.25%**

**252.** Ratio of Vodka and Wine in vessel A is 5 : 3 and same mixture in vessel B in the ratio of 3 : 2, 16l of mixture from vessel A taken out and poured in vessel B new ratio of Vodka to wine becomes 29 : 19. If new quantity of mixture in vessel B is equal to initial quantity of mixture in vessel A, then find quantity of Vodka after 16 l of mixture has been taken out from vessel A ?

- (a) 50l
- (b) 48l
- (c) 54l
- (d) 80l
- (e) 84l

**Answer (a) 50l**

**253.** Sameer and Veer entered into partnership business with the capital of Rs. x and Rs. (x + 12000) respectively. After One year Divyaraj joined them with capital of Rs. (x + 8000). At the end of 2nd year Sameer and Divyaraj withdraw their capital and Veer invested for one more year. If Sameer, Divyaraj and Veer gets profit in the ratio of 8 : 6 : 21 respectively. Find sum of capital invested by all three ?

- (a) 66000 Rs
- (b) 54000 Rs
- (c) 64000 Rs
- (d) 68000 Rs
- (e) 70000 Rs

**Answer (d) 68000 Rs**

**254.** Three vessels A, B and C of same capacity are all full. Vessel A and B contain mixture of milk and water in ratios 4 : 1 and 7 : 3 while vessel C contains only pure milk. The contents of all the vessels are poured into a larger vessel D. After drawing 30 liters of the mixture from vessel D, the final quantity of water in Vessel D is 40 liters. Find the capacity of each vessel

- (a) 80
- (b) 85
- (c) 95
- (d) 70
- (e) 90

**Answer (e) 90**

**255.** Two container P and Q have mixture. In P three liquid A, B and C is mixed in a ratio of 2 : 1 : 1 and in Q two liquid B and C is mixed in ratio of 3 : 2. Mixture of both container P and Q poured into third container X which show the quantity ratio of A, B and C is 8 : 13 : 10. Find the quantity in Q is what % less or more than the quantity in P.

- (a) 3.25%
- (b) 12.5%
- (c) 6.25%
- (d) 11%
- (e) 25%

**Answer (c) 6.25%**

**256.** Three container (A, B and C) have capacity in the ratio of 6 : 8 : 7 respectively. These three container are pour with three type of liquid X, Y and Z in the ratio 2 : 3 : 1 (in A), 1 : 2 : 5 (in B) and 3 : 1 : 3 (in C) respectively. Equal quantity of mixture is taken out from these containers and poured in container D. Quantity of liquid Y is  $\frac{1}{8}$  liter more than quantity of liquid X in container D. Find out the quantity taken out from each container.

- (a) 24
- (b) 35
- (c) 21
- (d) 20
- (e) None of these

**Answer (c) 21**

**257.** Amount invested in business respectively by Rana, Nikhil and Yash, was in AP respectively, and Yash invested thrice amount as that of Rana. Rana was manager and Nikhil was a programmer, and they both get salary of 8% and 5% of total profit respectively. If Rana invested for a month, Nikhil for 8 months and Yash invested for 4 months and Nikhil got Rs.6000 more than both of them together, find their total profit.

- (a) Rs. 100000/
- (b) Rs. 150000/
- (c) Rs. 174000/
- (d) Rs. 180000/
- (e) Rs. 200000/

**Answer (a) Rs. 100000/**

**258.** There are three vessels A, B & C having Milk to water ratio as 3 : 5, 1 : 3 and 7 : y. After adding mixture of vessel B to A, the milk to water ratio becomes 1 : 2 in A . Now 60% of mixture is taken out from vessel A and mixture of vessel C is mixed with remaining mixture of vessel A which results in final Milk to Water ratio of 3 : 4. Find the value of y given that volume of B is 20 ltrs and volume of C is (7 + y) ltrs.

- (a) 1
- (b) 4
- (c) 5
- (d) 3
- (e) 2

**Answer (b) 4**

**259.** There are three mixtures A, B and C. Mixture A contains 10% of water and B contains 15% of water. If A and B are mixed together than resultant mixture contains 12% water. while If B is mixed with C, which contains 22% water so the resultant mixture contains of 18% water. All these mixture are mixed in a jar and then 30 ml water is also added. Find total volume of water preset in jar, if initially mixture C contains 66 ml water.

- (a) 144 ml
- (b) 216 ml
- (c) 184 ml
- (d) 196 ml
- (e) 200 ml

**Answer (b) 216 ml**



**260.** There are two containers A and B filled with oil with different prices and their volumes are 140 litres and 60 litres respectively. Equal quantities are drawn from both A and B in such a manner that the oil drawn from A is poured into B and the oil drawn from B is poured into A. The price per litre becomes equal in both A and B. How much oil is drawn from each of A and B:

- (a) 40 litre
- (b) 80 litre
- (c) 30 litre
- (d) 21 litre
- (e) 42 litre

**Answer (e) 42 litre**

**261.** Vessel A, B and C contains mixture of milk and water. Ratio of the quantity of A, B and C is 25 : 20 : 18 respectively. In vessel A milk to water ratio is 2 : 3 and in vessel B water to milk ratio is 2 : 3. If vessel C contain milk to water ratio is 4 : 5 and all the mixture of 3 vessel poured to another vessel D then in vessel D quantity of water exceeds by 6 litres than the quantity of milk in that vessel. Find out the quantity of mixture contained in vessel A, B and C.

- (a) 50 Ltr., 40 Ltr., 36 Ltr
- (c) 75 Ltr., 60 Ltr., 54 Ltr
- (e) None of these
- (b) 44 4
- (d) 40 9 %

**Answer (a) 50 Ltr., 40 Ltr., 36 Ltr**

**262.** There are three fruit sellers Ramesh, Suresh and Vikash, all of them have a mixture of two types of apple. Ramesh has  $(x+18)$  kg of apple, Suresh has 12 kg of apple more than that of Ramesh while Vikash has 3 kg of apple more than that of Suresh. The two types of Apple are mixed in the ratio of 5 : 3, 2 : 3 and 4 : 3 for them respectively. If Ramesh sold 16 kg of mixed apple and added 12 kg of second type of apple then the new ratio becomes 5 : 6 for him. Vikash added y kg of second type of apple to his mixture and thus the new ratio of both types for him is 1 : 1. Then find the value of y.

- (a) 8 kg
- (b) 9 kg
- (d) 6 kg
- (e) 7 kg
- (c) 48 9 % 4 9 % Publications@

**Answer (b) 9 kg**

**263.** There are two vessels, vessel A and vessel B. Vessel A contains milk and water in the ratio 5 : 4 and vessel B contains Y litre of wine. 18 litre of mixture from vessel A is taken out and poured in vessel B. Now, 10 litre of mixture from vessel B is taken out and poured in vessel A. If quantity of wine in vessel A is found to be 4 litre then, find the initial quantity of wine in vessel B.

- (a) 8 L
- (b) 9 L
- (c) 10 L
- (d) 12 L
- (e) 6 L

**Answer (d) 12 L**

**264.** Three partners A, B and C invested their amounts in ratio of 2 : 5 : 8. At the end of four months, A invests some amount such that, his total investment will be equal to C's initial investment and C withdraw some amount that its total investment will be equal to B's initial investment. If 10% of B's share in profit of one year is Rs 2550. Then share of A and B in total annual profit will be?

- (a) 81500
- (b) 61200
- (c) 56100

- (d) 75000
- (e) None of these

**Answer (c) 56100**

**265.** A and B enter into a partnership business and decided to distribute 60% of profit as per (investment $\times$  time) ratio and rest in the ratio of 7:10. If the amount invested by A is 25 % more than B and B withdraw half of his total investment after 9 months. Find the profit distribution ratio of A and B according to given condition at the end of a year.

- (a) 43:41
- (b) 44:41
- (c) 48:43
- (d) 49:47
- (e) None of these

**Answer (b) 44:41**

**DIRECTIONS (Q. 266-267)**

Three partners Abhishek, Neeraj and Aniket enter into a partnership business with the capitals in the ratio of 3 : 5 : 8. After four month Abhishek, Neeraj and Aniket added Rs. 6000, Rs. 9000 and Rs. 12000 respectively, after next four months Abhishek and Aniket withdraw Rs. 5000 and Rs. 8000 respectively and Neeraj added additional Rs. 6000.

**266.** Two persons Ankur and Saurabh enter a business. Ankur invested capital 6000 more than what Abhishek invested for first four month and Saurabh invested what Aniket invested for last four months. If Ankur and Saurabh got profit share after one year in the ratio of 15 :28 then find the investment of Neeraj for last four months?(1 marks)

- (a) 24000 Rs
- (b) 18000 Rs
- (c) 16000 Rs
- (d) 30000 Rs
- (e) 36000 Rs

**Answer (d) 30000 Rs**

**267.** Amir and Mayank started a business in partnership. Amir investment for six months is Rs. 6000 more than what Neeraj invested for first four months and Mayank invested for eight months, Rs. 2000 less than what Aniket invested for first four months. If Mayank got profit share of Rs. 15400, out of total profit Rs. 22750 then find the investment of Amir is how much less than investment of Mayank? (2 marks)

- (a) 218000 Rs
- (b) 216000 Rs
- (c) 210000 Rs
- (d) 220000 Rs
- (e) 232000 Rs

**Answer (e) 232000 Rs**

**268.** Veer, Sameer and Gopal enter into a business by making investment in the ratio of 5 : 6 : 7 respectively. After six months Veer and Sameer 1 withdraw th and 25% of his initial investment respectively. After 5 1 eight months from the starting of business Gopal added 7 th of his initial investment. If at the end of one year profit share of Gopal is Rs. 2000 more than Sameer, then find profit share of Veer?

- (a) 4420 Rs
- (b) 4520 Rs
- (c) 4820 Rs
- (d) 4320 Rs
- (e) 4720 Rs

**Answer (d) 4320 Rs**

- 269.** Manish and Rituraj invested Rs. 12000 and Rs. 16000 in a business. After four months Manish and Rituraj both added Rs. 4000 in their initial investment. At the end of one year the total profit was Rs. 172500, if Manish and Rituraj invested their profit share on compound interest at the rate of 20% and 10% respectively then find difference between interests got by both at the end of two years?

- (a) Rs.10250
- (b) Rs.11520
- (c) Rs.12210
- (d) Rs.13110
- (e) Rs.12660

**Answer (d) Rs.13110**

**DIRECTIONS (Q. 270-271)**

Data given below about investment of three partners in a business. Read the data carefully and answer the questions: Swati, Monika and Anshika three business partners enter into a business by making investment Rs. 4000, Rs. 6000 and Rs. 8000 respectively. After first quarter Swati, Monika and Anshika added some amount in the ratio of 2 : 3 : 2 1 5. After second quarter Swati, Monika and Anshika withdrew 5<sup>th</sup>, 3<sup>rd</sup> and 3 7<sup>th</sup> respectively of capital invested in second quarter. At the end of nine month profit share divided among Swati, Monika and Anshika in the ratio of 24 : 37 : 49 respectively.

- 270.** Satish and Veer enter into a business, Satish invested 200% more than Swati capital for second quarter, while Veer invested 300% more than Anshika capital for third quarter. And profit share of Satish at end of two year is Rs. 7250 out of total profit of Rs. 15950. Then find for how many months Veer invested his capital?

- (a) 14 months
- (b) 12 months
- (c) 18 months
- (d) 10 months
- (e) 8 months

**Answer (c) 18 months**

- 271.** Monika and Anshika enter into a business with another partner Shikha. Monika invested 144% of what she invested in second quarter, while Anshika invested 160% of what she invested in second quarter. If all three invested their capital for a year and Shikha gets a profit of Rs. 9600 out of total profit of Rs. 28000, then find investment of Shikha?

- (a) 12,800 Rs
- (b) 11,600 Rs
- (c) 14,400 Rs
- (d) 14,200 Rs
- (e) 13,600 Rs

**Answer (c) 14,400 Rs**

- 272.** Three partners A, B and C invested in the ratio of 4 : 5 : 5 in a business. After 3 months A increased his capital by 40%, after 3 months more A again increased his capital by 20%. B increased his capital by 25% after every 4 months. C increased his capital by 25% after 3 months and after 4 months more C increased his capital by 20%. If the total profit of Rs. 30,130 is earned at the end of the year, find what was the A's share of profit?

- (a) Rs. 12,420
- (b) Rs. 14,420
- (c) Rs. 13,240
- (d) Rs. 15,340
- (e) Rs. 12,340 8

**Answer (a) Rs. 12,420**

- 273.** Veer, Sameer and Satish entered into a business by investing equal amount each. After four months Sameer added half of his initial investment, while Veer and Satish withdrew half of their initial investment. At end of one year the total profit is Rs. 8000 and Veer and Sameer invested their profit share at the rate of 10% compounded annually for some time in a scheme. If the difference of the amount received by Veer and Sameer from the scheme at the end is Rs. 2420, find for how much time Veer and Sameer invested in the scheme? 1

- (a) 4 yr
- (b) 3 2 yr
- (c) 3 yr
- (d) 2 yr
- (e) 1yr

**Answer (d) 2 yr**

- 274.** P, Q and R entered into a partnership with their initial capital in the ratio of 3 : 5 : 8 respectively. After third month P withdrew 3rd of his initial capital and Q withdrew half of his initial capital. After next six months P increased his capital by 100% more, Q increased by 300% more and R increased by 25% more. If at the end of an year, difference between profit share of Q and R was Rs. 7000, then find profit share of P?

- (a) Rs. 6600
- (b) Rs. 7700
- (c) Rs. 5500
- (d) Rs. 5750
- (e) Rs. 5950

**Answer (c) Rs. 5500**

- 275.** A, B & C, three friends started the business in which A invested for 6 months, B initially didn't invest & started as working partner, while C invested for 4 months. They decided to donate 1% of total profit and to give 22% of total profit to B as salary. Find the ratio of their profit shares (A:B:C) if A and C invested in the ratio of 2 : 3 and B also invested an amount which is 80% of total amount invested by A and C together for a single month.

- (a) 1 : 3 : 1
- (b) 1 : 1 : 3
- (c) 3 : 1 : 3
- (d) 1 : 1 : 1
- (e) None of these

**Answer (d) 1 : 1 : 1**

- 276.** Sweta and Neha profess to tell their present ages as 25 and 20 years respectively. (Not original age). Ratio of their original ages 5 year ago is 5 : 4. Sum of ages of both 5 years hence is 9 % more than the sum of present ages of both professed by them. Find the ratio of their present original age.

- (a) 4:7
- (b) 5:6
- (c) 6:5
- (d) 4:5
- (e) 5:7

**Answer (e) 5:7**

**277.** Average age of a man, woman and their son is 30 years. Man's age is 1 two year more than his wife and age of son is 4th the sum of age of his mother and father. When two other family members were added, new average becomes 27 years. If difference between age of two new member are one year then find difference between son and the new member who is elder.

- (a) 7 years
- (b) 8 years
- (c) 4 years
- (d) 2 years
- (e) 5 years

**Answer (e) 5 years**

**DIRECTIONS (Q. 278-279)**

Data given below about investment of three partners in a business. Read the data carefully and answer the questions: Swati, Monika and Anshika three business partners enter into a business by making investment Rs. 4000, Rs. 6000 and Rs. 8000 respectively. After first quarter Swati, Monika and Anshika added some amount in the ratio of 2 : 3 : 2 1 5. After second quarter Swati, Monika and Anshika withdrew 5 th , 3 rd and 3 7 th respectively of capital invested in second quarter. At the end of nine month profit share divided among Swati, Monika and Anshika in the ratio of 24 : 37 : 49 respectively.

**278.** There are some teachers in a college with average age of 45 years. Sum of maximum age and minimum age is 108 years and difference of maximum age and minimum age is 27 years. If fourteen teachers left the college with the average age of 32.5 years and six new teachers joined college. Out of six new teachers four ages are equal to maximum age teacher and two teachers age equal to minimum age teacher. If new average becomes 49 teachers initially?

- (a) 54
- (b) 60
- (d) 68
- (e) 72 10
- (c) 35 years 1 1 1 3 : 2 : 4. 3 8 th 4 7 years, then find the number of Publications@

**Answer (c) 35 years 1 1 1 3 : 2 : 4. 3 8 th 4 7 years, then find the number of Publications@**

**279.** The ratio between present age of Veer & Saurbh is 3: 4 and between present age of Divyaraj & Gopal is 2: 3. If sum of present age of Veer, Saurbh, Divyaraj & Gopal is 128 years and ratio between sum of age of Veer & Saurbh to sum of age of Divyaraj & Gopal is 63: 65, then find the sum of age of Veer & Gopal after eight years.

- (a) 90 years
- (b) 78 years
- (c) 86 years
- (d) 84 years
- (e) 82 years

**Answer (e) 82 years**

**280.** 2n years ago, the age of Raju was four times that of his son and n years ago, the age of Raju was thrice that of his son. If n years later, the sum of the ages of Raju and his son will be 80 years, then the difference in the ages of Raju and his son is

- (a) 20 years
- (b) 40 years
- (c) 24 years
- (d) 30 years
- (e) 34 years

**Answer (d) 30 years**

**281.** '2n' years ago ratio of Amit's age to Inder's age is 5 : 4. 'n' years ago ratio of Inder's age to Satish's age is 9 : 7. Difference between present age of Amit to Satish's present age is 12 years. Find the sum of present ages of all three if ratio of Amit's age to Satish's age after 'n' year will be 13 : 9.

- (a) 81
- (b) 84
- (c) 87
- (d) 90
- (e) Cannot be determined

**Answer (d) 90**

**282.** P1 is the series of four consecutive numbers divisible by 4 with an average of 34, while P2 is the series of five consecutive numbers divisible by 9, and whose second lowest number is equal to second highest number of P1 series. If P3 is the series of seven consecutive 1 numbers, whose lowest number is 3<sup>rd</sup> of lowest number of P2 series, then find the sum of average of P2 & P3 series?

- (a) 53
- (b) 51
- (c) 57
- (d) 55
- (e) 49 11

**Answer (c) 57**

**283.** Average of five two digit numbers, in which 2 numbers say A and B are reverse of each other, is 25.4. If all three numbers except A and B multiplied by (-2) then the average of five numbers decreased by 43.2. What could be the greatest value among A and B.

- (a) 32
- (b) 23
- (c) 50
- (d) 41
- (e) 14

**Answer (d) 41**

**284.** Average of any 200 consecutive natural numbers is 499.5. If next 1000 numbers more add in it then find the new average.

- (a) 1035.5
- (b) 1299.5
- (c) 1199.5
- (d) 1099.5
- (e) 999.5

**Answer (e) 999.5**

**285.** Average age of a group of people is four times of the number of people in the group. Sakshi leaves the group and the average age is still four times of the number of people in the group. After that Sheetal leaves the group and the average age is still four times of the number of people in the group. If ratio between Sakshi's age to Sheetal's age is 21 : 19, then find the average age of the group if Ritu leaves the group whose age is 20 years

- (a) 36 years
- (b) 37 years
- (c) 38 years
- (d) 39 years
- (e) 40 years

**Answer (c) 38 years**

**DIRECTIONS (Q. 286-287)**

Rakesh distributed Rs.50000 into (X, Y, Z) 3 groups having 50 people in total. Number of people in group 'X' is equal to the number of people in group 'Z' and total money got by group 'Y' is equal to the total money got by group 'Z'. Average money got by group X and Y 3500 together is Rs. 3 and average money got by group Y and Z is Rs.1000

**286.** Find the number of people in group Y?

- (a) 20
- (b) 16
- (c) 10
- (d) 18
- (e) 24

**Answer (c) 10**

**287.** Find the amount got by each person in group Z.

- (a) 800 Rs
- (b) 600 Rs
- (c) 1000 Rs
- (d) 750 Rs
- (e) 500 Rs

**Answer (d) 750 Rs**

**288.** A publishing house started its publication with the 1500 DI books. Printing cost per book is 125 Rs, packing cost per book is 15 Rs and shipping cost per book is 40 Rs. if 50% of books sold on half of total cost price, then find on how much percent above should be remaining book sold to gets 20% profit on total outlay?

- (a) 50%
- (b) 60%
- (c) 80%
- (d) 90%
- (e) 70%

**Answer (d) 90%**

**289.** A publishing house sold a puzzle book in Rs. 475 and made profit of 25% on CP and sold a DI book in Rs. 575 and made a profit of 15% on CP. If publication sold puzzle book in Rs. 360, then find what price should DI book will be sold to made a profit of 30% on both the books ?

- (a) Rs.784
- (b) Rs.874
- (c) Rs.847
- (d) Rs.748
- (e) Rs.478 6

**Answer (a) Rs.784**

**DIRECTIONS (Q. 290)**

Abhishek bought some chairs and tables from a shopkeeper. The marked price of a chair and a table were in the ratio 5 : 8. The shopkeeper gave discounts of 20% and 25% on the chair & the table respectively. The ratio of number of chairs and tables bought by Abhishek is 6 : 5. 15. If Abhishek sells each chair and table bought by him at discounts of 25% and 20% respectively after marking up the prices of both by 50% and gives one table free for every four chairs bought by a customer 2 and only 3<sup>rd</sup> of the total chairs are sold in bunch of four chairs, then what is the net profit /loss % made by Abhishek after selling all of the items which he bought from the shopkeeper?

- 290.** If the marked price of a table set by the shopkeeper was Rs.300 more than that of a chair and the total expenditure made by Abhishek in purchasing the chairs and table from the shopkeeper was Rs.108000, then how many chairs were purchased by Abhishek?

- (a) 150
- (b) 60
- (c) 120
- (d) 90
- (e) 80

**Answer (c) 120**

**DIRECTIONS (Q. 291-292)**

A car 'X' is manufactured in a company 'A'. Cost of raw material on a car is 25% of total cost and cost of labor is 20% of rest. 30% of total cost occur due to four engineers working there and remaining cost is due to machines and taxes which is in the ratio 7 : 3. Company sold the car 'X' at 28% profit.

- 291.** If cost of raw material is increased by 30% and all their 4 engineers leave the company, so they hire 2 new engineers by giving 50% more salary as they give to previous engineer each. Find the % change in selling price so overall profit must remain same.

- (a) 0%
- (b) 100%
- (c) 2%
- (d) 50%
- (e) 3%

**Answer (a) 0%**

- 292.** Company A, started making car 'Y' in which same raw material used, labor cost is increased by 20%. If all other cost is same then find the ratio of selling price of car 'X' to car 'Y', given that car 'Y' sold at 20% profit.

- (a) 331 : 309
- (b) 320 : 314
- (c) 313 : 309
- (d) 320 : 309
- (e) 309 : 320

**Answer (d) 320 : 309**

- 293.** Marked price of two articles A and B are in the ratio of 7 : 9, at the time of selling shopkeeper gives discount of d% on article A and (d + 5)% on article B and made a profit of 25% on each article, if cost price of article A and B are in ratio 112 : 135. Then find percent of discount given by shopkeeper on both articles?

- (a) 12.5%, 17.5%
- (b) 25%, 30%
- (c) 10%, 15%
- (d) 15 %, 20%
- (e) 20%, 25%

**Answer (e) 20%, 25%**



**294.** Neeraj purchased two mobile from a shop. He sold first mobile at the price of Rs. 18750 and second mobile at the price of Rs. 14250. If the profit percent on first mobile is five times of the loss percent on second mobile, then find the overall profit made by Neeraj after selling both the mobile phones. (C.P of both mobile is same)?

- (a) 15%
- (b) 12%
- (c) 14%
- (d) 10%
- (e) 18%

**Answer (d) 10%**

**295.** Ratio between marked price of article A to article B is 4 : 5. Shopkeeper allowed d% discount on article 'A' and (d + 18)% discount on article 'B', so selling price of both articles become equal. If shopkeeper made a profit of 20% on article A and 25% on article B and profit made on article B is Rs. 384 more than that of article A, then find the cost price of article 'A' and article 'B' respectively?

- (a) 9000 Rs. 8400 Rs
- (b) 9600 Rs. 9216 Rs
- (c) 9800 Rs. 9012 Rs
- (d) 9600 Rs. 8488 Rs
- (e) 9200 Rs. 9216 Rs

**Answer (b) 9600 Rs. 9216 Rs**

**296.** Aman started a shoes manufacturing company, manufacturing cost of each shoes is Rs. 500 , Raw material cost is Rs 300 and transport cost of Rs 100 If company manufactured 150 Shoes in first order and sold 50% of shoes on half of its cost price. Then find at what price of should the company sold the remaining shoes to gets 50% profit on its total cost?

- (a) 100%
- (b) 150%
- (c) 75%
- (d) 125%
- (e) 120%

**Answer (b) 150%**

**297.** The ratio between marked price of a same watch in two shops P and Q is 7 : 5 . In shop P watch available at two successive discounts of 10% & 25% , while in shop Q watch available at two successive discounts of d% and 10%. If selling price of watch at shop Q is less than that of 44 selling price of watch at shop P by 105 th of the selling price of watch at shop P, then find the value of 'd' ?

- (a) 39%
- (b) 29%
- (c) 31%
- (d) 35%
- (e) 37%

**Answer (a) 39%**

**298.** Cost price of a Samsung mobile is 20% more than cost price of a Mi mobile. Shopkeeper marked up Samsung mobile at 25% above its cost price and Mi mobile at 40% above cost price. If shopkeeper allowed discount of 20% on Mi mobile and 25% on Samsung mobile and difference between selling price of Samsung mobile and Mi mobile was 160 Rs. then find sum of marked price of one Samsung mobile and one Mi mobile?

- (a) 92840 Rs
- (b) 92260 Rs
- (c) 98460 Rs
- (d) 92800 Rs

(e) 98530 Rs

**Answer (d) 92800 Rs**

**299.** In a city 60 percent are male and remaining are females. Out of total males, 20% of male are government employees. Out of remaining males, ratio between private employees to unemployed male is 5 : 7. Out of total females, 40% of female are in government jobs and out of remaining females 40% of female work for private firms and remaining are unemployed. Find the total number of male and female, who are government employees, if difference between male and female who are unemployed is 3400?

- (a) 6000
- (b) 5000
- (c) 7000
- (d) 4000
- (e) 3000

**Answer (c) 7000**

**300.** A person 'A', invested  $37\frac{1}{2}\%$  of his savings in scheme XYZ, while B invested  $62\frac{1}{2}\%$  of his savings in the same scheme. If B invested 50% of his remaining savings in scheme PQR, and A thought to invest double of the amount invested by B in PQR scheme then find what percent of A's remaining savings after investment in XYZ, A should invest in PQR. Given that amount invested by B in XYZ is twice the amount invested by A in same scheme.

- (a) 80%
- (b) 68%
- (c) 64%
- (d) 72%
- (e) 60%

**Answer (d) 72%**

**301.** A man bought few kg's of sugar of 3 variety (X, Y, Z) in the ratio of their quantity 2 : 3 : 5 respectively and started to sell them at profit of 20%, loss of 10% and profit of 32% respectively. He managed to sell only 80% of total sugar and rest was return at the same price as he bought. The price of these sugar is Rs. 30/kg, Rs. 40/kg and Rs. 50/kg respectively. Calculate his total profit amount, if total sugar bought was 50 kg and he sold all the sugar of X and Y types.

- (a) 320
- (b) 300
- (c) 0
- (d) 180
- (e) None of these

**Answer (e) None of these**

**302.** In an election survey,  $83\frac{1}{3}\%$  of total voters took part in survey, 50% then claims to votes for candidate A, 10% are uncertain & rest says to vote for B. If all of them voted according to their commitments on the day of election and those who were not part of survey, voted to A and B in ratio of 2 : 1. All the people who are uncertain in survey, voted to A : B in the ratio of 1 : 4 in election. If A won by 640 votes then find total number of votes in election.

- (a) 6000
- (b) 5000
- (c) 9000
- (d) 7200
- (e) 6750 2

**Answer (d) 7200**

**303.** In 2016 a store sold some mobile phone at certain price. In 2017 store increase selling of mobile phone by reducing the price of mobile phone by 25%. The total revenue generated by selling mobile phone in 2017 is increased by 5% compared to previous year. The total revenue generated by store in 2017 is 94.5 lakh and number of mobile phones sold by store in the year 2017 is 90 more than that in 2016. Find initial cost of one mobile.

- (a) 32000 Rs
- (b) 40000 Rs
- (c) 36000 Rs
- (d) 24000 Rs
- (e) 20000 Rs

**Answer (b) 40000 Rs**

**304.** Veer bought some Shirts & some Jeans from a store. The marked price of a shirt and a jeans is in the ratio 5 : 7 and store allows discounts of 20% and 25% on the shirt & the jeans respectively. The ratio between number of shirt and jeans bought by Veer is 9 : 8. Veer marking up the prices of both by 50% on the price at which he bought and sells each shirt and jeans bought by him at discounts of 25% and 20% respectively. If Veer gives one 1 shirt free for every 4 jeans bought by a customer, then what is the net profit made by Veer after selling all of the items which he bought from the store?

- (a) 2%
- (b) 7%
- (c) 6%
- (d) 5%
- (e) 8%

**Answer (d) 5%**

**305.** In a store marked price of a shirt is 33 jeans. There are two discounts schemes A & B. In Scheme A, if someone buy one jean and one shirt together, then store gives overall discount of 35%, while in scheme B if someone buy one shirt and one jeans separately, store give 20% discount on shirt and 25% on jeans. If difference between selling price of one jeans and one shirt in scheme A and B is Rs. 384 then find cost price of one jeans & one shirt, given that store made of 33 sold one jeans and one shirt in scheme B ?

- (a) 720 Rs, 1150 Rs
- (c) 786 Rs. 1168 Rs
- (e) 790 Rs. 1108 Rs
- (b) 21 17
- (d) 23 23 %

**Answer (b) 21 17**

**306.** Veer invested Rs 22500 for 2 year at the Rate of x% in scheme A at compound interest annually and gets a total amount of Rs 32400 If he added Rs 2600 in amount and invested total amount in scheme B at S.I. for 3 year at same rate . Then find the total simple interest veer gets from scheme B?

- (a) Rs 22500
- (b) Rs 22000
- (c) Rs 17500
- (d) Rs 20000
- (e) Rs 21000

**Answer (e) Rs 21000**

**307.** Satish invested 16000 Rs. in simple interest for 2 years on certain rate and gets an interest of 4800 Rs, if he invested total amount (Principle + Interest) in a scheme, which offered compound interest on 5% more interest rate as earlier rate. Then find total interest gets by Satish after 2 years ?

- (a) 9252 Rs
- (b) 9225 Rs
- (c) 9512 Rs
- (d) 925 Rs
- (e) 9152 Rs

**Answer (e) 9152 Rs**

**308.** Bhavya and Veer invested their principle in two different schemes, Bhavya invested X Rs. on compound interest for two year at rate of 20% annually and Veer invested 4000 Rs. more than Bhavya on simple interest for three year at 15% annually, if both gets total interest of Rs. 9632, then Find the amount invested by Veer ?

- (a) 12900
- (b) 12400
- (c) 8800
- (d) 12800
- (e) 12600

**Answer (d) 12800**

**309.** A man invested Rs. x, y and z in three difference schemes which offers 10% p.a., 20% p.a. and 30% p.a. rate respectively and after one year he earns Rs. 2000 interest. If x, y and z are in arithmetic progression and sum of x, y and z is Rs. 9,000. Find the difference between 'x' and 'z'.

- (a) 1500
- (b) 2000
- (c) 2500
- (d) 2250
- (e) 1750

**Answer (b) 2000**

**310.** P invested Rs. X in a scheme for 2 year which offered simple at the rate of 15% per annum and Q invested Rs. (X + 2500) in another scheme for same period of time, which offered compound interest at the rate of 20% per annum. If from both scheme P and Q got total interest of Rs. 32550, then find the value of X ?

- (a) 41500
- (b) 42500
- (c) 40500
- (d) 40000
- (e) 38250

**Answer (b) 42500**

**311.** Rajat have some money in his hand. He invested 20% of the money in scheme 'A' for 4 year at 6% p.a., 30% of the amount in scheme 'B' for 6 year at 12% p.a., remaining in scheme 'C' 2 year at 15% p.a. If total amount Rajat received from scheme 'A', 'B' and 'C' is 11,355, then find the difference of sum invested in scheme 'B' and scheme 'A'.

- (a) 1800
- (b) 1500
- (c) 1200
- (d) 750
- (e) 270 9

**Answer (d) 750**

- 312.** Rahul took a loan of Rs.80000/- at rate of 5% p.a. at SI & lent it at 5% pa at CI. After 3 years, he cleared all his debts and invested the profit earned in a scheme which offers C.I. After two years he got Rs.128.1/- as interest from the scheme. Find the rate of interest offered by scheme.
- (a) 8%
  - (b) 7%
  - (c) 6%
  - (d) 10%
  - (e) 12%

**Answer (d) 10%**

- 313.** A man invested his total savings in two parts. He invested Rs. 30000/- in a scheme which offers CI at ROI of 10% per annum and remaining in SI at ROI of 5% per half year. After two years the amount obtained by him in former scheme is half of later, then calculate amount invested in later scheme.
- (a) Rs. 242000
  - (b) Rs. 60500
  - (c) Rs. 12100
  - (d) Rs. 80525
  - (e) None of these

**Answer (b) Rs. 60500**

- 314.** Satish saved 55% of his monthly income, which he further invested in two different schemes A & B in the ratio of 7 : 5 on CI compounded annually at the rate of 10% p.a. and 20% p.a. respectively. If after two years, Satish gets total interest of Rs. 8074 from both the schemes, then find the total annual saving of Satish?
- (a) 316,600 Rs
  - (b) 316,400 Rs
  - (c) 316,000 Rs
  - (d) 316,200 Rs
  - (e) 316,800 Rs

**Answer (e) 316,800 Rs**

- 315.** Aman invest Rs.  $(x - 2000)$  in scheme 'P' which offers 20% CI pa while Rs.  $(x + 2000)$  in scheme 'Q' which offers 8% SI pa. After 2 year difference between interest earn from scheme P and Q is 1600, then amount invested by Aman in scheme 'Q' is what percent more/less than amount invested by him in scheme 'P' ? 1
- (a) 75%
  - (b) 133 3 %
  - (c) 150%
  - (d) 50%
  - (e) 33 3 %

**Answer (d) 50%**

- 316.** Ritu invested her total saving in three different FD schemes A, B and C in the ratio of 5 : 4 : 6 on CI for two years at the rate of 10%, 15% and 20% respectively. If interest is calculated annually and interest from scheme B is Rs. 744 more than interest from scheme A then, find difference between interest received from scheme C and scheme B by Ritu?
- (a) Rs. 4185
  - (b) Rs. 4175
  - (c) Rs. 3840
  - (d) Rs. 4580
  - (e) Rs. 3250

**Answer (a) Rs. 4185**

**DIRECTIONS (Q. 317)**

A person invested Rs. 20000 in a bank which is offering 10% per annum simple interest. After two years he withdrew the money from the bank and deposited the total amount in another bank which gives an interest rate of  $r\%$  p.a. compounded annually. After 2 years he received an amount of Rs. 2460 more than what he had invested in that bank.

**317.** If the person had invested Rs. 50,000 instead of 20000 in the bank that offered simple interest, what would have been his net profit after following the same procedure as given above?

- (a) Rs. 16,800
- (b) Rs. 16,150
- (c) Rs. 16,350
- (d) Rs. 16,000
- (e) None of these

**Answer (b) Rs. 16,150**

**318.** A bank offers 10% p.a. at CI in scheme 'A', 20% p.a. at CI in scheme 'B' and 40% p.a. at CI in scheme 'C'. Veer have same amount in his hand. He invested 32% amount in scheme 'A', 20% in scheme 'B' and remaining in scheme 'C'. After two year he received 6600 as interest. Find the difference between amount invested by Veer on scheme 'C' to that of in scheme 'B'.

- (a) 5,000
- (b) 4,000
- (c) 2,000
- (d) 3,000
- (e) 2,500

**Answer (d) 3,000**

**319.** Bhavya borrowed Rs. 37500 on compound interest with agreement that, he will not pay amount after 2 year with interest then the rate of interest increase by 5% as earlier interest rate. If Bhavya paid Rs. 12093.75 as total compound interest after two year. Then find how much amount Bhavya have to pay after two year for completing his entire borrowing?

- (a) Rs. 58000
- (b) Rs. 52000
- (c) Rs. 42000
- (d) Rs. 45000
- (e) Rs. 54000 12

**Answer (e) Rs. 54000 12**

**320.** A man invested 'X' amount in scheme A and 'Y' amount in scheme B. Scheme A offers 8% p.a. at S.I. scheme B offer 20% p.a. at C.I. Man invested both schemes for 2 years. The sum of interest he earns after 2 years will be equal to the simple interest he can earn if he invested 'Y' amount in scheme A for 7 years. Find the total amount he has initially if difference between interest earned by him after 2 years is 2304.

- (a) 12,300
- (b) 12,900
- (c) 12,600
- (d) 13,500
- (e) 13,800

**Answer (c) 12,600**

**321.** Veer invested Rs.  $x$  in SBI for three years on S.I. at the rate of 15% per annum and Sameer invested Rs.  $(x + 4000)$  in BOB for two years on C.I. at the rate of 8% per annum. If Divyaraj invested equal to sum of what Veer and Sameer invested in UBI for two years on C.I. at the rate of 20% per annum. Find the sum invested by Divyaraj, if they all got total interest of Rs. 20382.4?

- (a) Rs. 20000
- (b) Rs. 18000
- (c) Rs. 28000
- (d) Rs. 30000
- (e) Rs. 26000

**Answer (c) Rs. 28000**

**322.** Three employee Ankit, Veer and Sameer got Rs.  $P$ , Rs.  $(P+2400)$  and Rs.  $(P+4400)$  Diwali bonus. Ankit and Veer invested their half of bonus on CI at the rate of 10% and 20% respectively for two years. Sameer invested 60% of his bonus on simple interest at the rate of 15% p.a. for three years. If Sameer got Rs. 132 more as interest got by Ankit and Veer together, then find the bonus got by Sameer?

- (a) 12000 Rs
- (b) 10000 Rs
- (c) 14400 Rs
- (d) 14000 Rs
- (e) 18000 Rs

**Answer (d) 14000 Rs**

**323.** Manoj lend Rs.  $P$  for three years on S.I. at the rate of 15% per annum and Rajesh lend Rs.  $(P + 8000)$  for two years on C.I. at the rate of 8% per annum. Suresh borrowed sum equal to of what Manoj and Rajesh lend, for two years on C.I. at the rate of 20% per annum. If Suresh paid interest Rs. 5352 more than, what Manoj and Rajesh got total interest on their sums together. Find total sum borrowed by Suresh?

- (a) Rs. 34000
- (b) Rs. 44000
- (c) Rs. 32000
- (d) Rs. 46000
- (e) Rs. 30000

**Answer (c) Rs. 32000**

**324.** A person invested his savings of Rs. 32000 in two parts at equal rate of interest in CI and SI. After 3 years, he withdrew amount invested in CI, and put the cash in his almirah. After another 2 years, the amount invested in SI got doubled and he also withdrew them. If the final ratio of both the amount is 36:65, then find amount invested by him in SI (in Rs.)

- (a) 20000/
- (b) 18050/
- (c) 19500/
- (d) 21500/
- (e) can't be determined

**Answer (c) 19500/**

**325.** A man invested an amount into three schemes P, Q and R in the ratio of 5 : 6 : 9 at the rates of 20%, 15% and 10% respectively. Schemes P and R offered compound interest annually, while scheme Q offered simple interest annually. Total interest received by man from scheme P and Q together is Rs. 1899 more than total interest received from scheme R after two years. If man would invest total amount on C.I at the rate of 15% p.a., then find the total interest received by the man?

- (a) 5800 Rs
- (b) 5805 Rs
- (c) 5580 Rs

- (d) 5590 Rs
- (e) 5900 Rs

**Answer (b) 5805 Rs**

**326.** Point A to Point B is a downstream journey of 300 km on a stream which flows at a speed of 5 km/hr. Two boats P and Q starts from point A and Point B respectively with speed of 25 km/hr and 15 km/hr in still water. After reaching the opposite point they return to their starting points, find after how much time will they meet second time?

- (a) 7.5hour
- (b) 15hour
- (c) 20hour
- (d) 10hour
- (e) None of these

**Answer (c) 20hour**

**327.** Distance between Delhi and Jaipur is 300 km. Aman starts from Delhi and Rajiv from Jaipur at same time. After two hours, Aman realized he was travelling slow and therefore increased his speed by 25% and meet Rajiv at a point 108 km from Delhi. Find the increased speed of Aman, if Rajiv derived at a constant speed of 75 km/hr.

- (a) 40 km/hr
- (b) 50 km/hr
- (c) 60 km/hr
- (d) 55 km/hr
- (e) 65 km/hr

**Answer (b) 50 km/hr**

**DIRECTIONS (Q. 328-329)**

Time taken by train A to cover a distance is 1.5 times the time taken by train B to cover the same distance. Train A and train C starts simultaneously from a station in same direction. Train B, start after 30 min and overtake train C in 1.5 hour later than it cross train A.

**328.** If speed of train A is 80 km/hr, then what is the speed of train C. (2 Marks)

- (a) 60 km/hr
- (b) 100 km/hr
- (c) 125 km/hr
- (d) 75 km/hr
- (e) 80 km/hr

**Answer (b) 100 km/hr**

**329.** If train A double its speed, then speed of train A is what % more than the speed of train C. (1 Mark)

- (a) 60%
- (b) 100%
- (c) 80%
- (d) 50%
- (e) None of these

**Answer (a) 60%**



**DIRECTIONS (Q. 330-331)**

A consignment should deliver on time to customer for which Mahendra start his journey with uniform speed, after 2 hours customer call Mahendra to deliver his order 1 hour before the decided time. Mahendra increased his speed by 50% to deliver it 1 hour before the decided time.

**330.** Find the total time taken by Mahendra to deliver the consignment?

- (a) 4 hours
- (b) 5 hours
- (c) 6 hours
- (d) 7 hours
- (e) 3 hours

**Answer (b) 5 hours**

**331.** If initially, customer also start moving toward Mahendra at speed of 60 km/hr and take his consignment after 3 hours then find the total distance between Mahendra and Customer initially.

- (a) 256 km
- (b) 150 km
- (c) 500 km
- (d) 450 km
- (e) 300 km

**Answer (d) 450 km**

**DIRECTIONS (Q. 332-333)**

A man decided to run 15 rounds of a circular track of 400 m in certain time with certain speed. He starts running but after completing some round around the track he reduced his speed by 40% due to which he takes 4 min extra as scheduled. But if he reduced his speed by completing 3 more rounds he would have reached 2 min earlier than the time he actually reached.

**332.** Find the original speed of man?

- (a) 240 m/min
- (b) 160 m/min
- (c) 200 m/min
- (d) 400 m/min
- (e) 350 m/min

**Answer (d) 400 m/min**

**333.** Find the number of rounds at which he decided to reduce his speed?

- (a) 7
- (b) 9
- (c) 10
- (d) 8
- (e) 11

**Answer (b) 9**

**334.** A cyclist left point A for point B and travelled at the constant speed of 25 km/h. When he covered the distance of  $25/3$  km, he was overtaken by a car that left point A twelve minutes after the cyclist and travelled at a constant speed too. When the cyclist travelled another 30 km, he encountered the car returning from B. Assume that the car did not stop at point B. Find the distance between A and B.

- (a) 39.5833 km
- (b) 41.0833 km
- (c) 60.833 km
- (d) 43.33 km
- (e) 50 km

**Answer (c) 60.833 km**

- 335.** Two person left simultaneously two places A and B. One of them left A for B while the other left B for A. Both travel at uniform speed . The first person on reaching B returns to A and then again travels back to B and so on. Similarly the second person on reaching A returns to B and then travels back to A and so on. What will be the distance covered by the first person when they meet for the third time given the ratio of the speed of the first person to that of the second person is 3 : 2 and the distance between A and B is 500 m?

- (a) 1000 m
- (b) 1500 m
- (c) 2500 m
- (d) 1200 m
- (e) 1800 m

**Answer (b) 1500 m**

- 336.** A train M running at the speed of 108 km/hr crosses a man running at 12 km/hr in opposite direction of train in 12 second and crosses a platform in 32 sec. If a train N is Standing on same platform and length of platform is 140 meter more than length of train N. Find in what time train N will cross train M running in same direction, if train N passes a pole in 12 second?

- (a) 168 sec
- (b) 164 sec
- (c) 154 sec
- (d) 186 sec
- (e) 172 sec

**Answer (b) 164 sec**

- 337.** A bus (X) start its journey from P to Q simultaneously another bus (Y) start from Q to P they reach their respective destination after 3 hour. Bus X travel with a speed of 40 km/hr in first hour 50 km/hr in second hour and 60 km/hr in 3rd hour. Bus Y travel with a speed of 60 km/hr in first hour, 40 km/hr in second and 50 km/hr in last hour. Find out the distance between Q and the point where the buses cross each other.

- (a) 87
- (b) 88
- (c) 89 2 7
- (d) 82 9
- (e) 87 9 175 2

**Answer (d) 82 9**

- 338.** Two trains P and Q started from two points A and B towards each other to cover 300 km distance. Speed of train Q is twice of speed of train P. If train P runs at its normal speed and train Q reduces its speed to 50% then it will take 2.5 hrs. more to meet each other. Find the time (in sec) in which train P can cover a platform of 150m length if the length of train P is given as 50m.

- (a) 18 seconds
- (b) 36 seconds
- (c) 72 seconds
- (d) 54 seconds
- (e) 24 seconds

**Answer (b) 36 seconds**

**339.** A train P, 180 meter long train passed a pole in 4 sec and also passed two trains Q and R in 9 sec and 39 sec respectively, where train Q running in opposite direction of train P and train R is running in same direction of train P. If length of train Q and R is 240 meter and 210 meter respectively, then in what time train Q will pass train R, if both runs in opposite direction ? 7 3

- (a) 35 sec
- (b) 9 11 sec
- (c) 12 11 sec
- (d) 15 sec
- (e) 55 sec

**Answer (c) 12 11 sec**

**340.** Total distance between Delhi to Lucknow is 480 km. A train starts running with an average speed of 60 km/hr from Delhi to Lucknow while another train starts Journey after 120 minutes of first train and reaches Lucknow 30 minutes before first train. If first train stops for 5 minutes on each station and second train did not stop at any station ,then find the ratio between speed of first train to speed of second train given that total number of station between Delhi and Lucknow are nine.?

- (a) 25 : 33
- (b) 25 : 32
- (c) 35 : 25
- (d) 25 : 31
- (e) 21 : 31 11

**Answer (b) 25 : 32**

**341.** The ratio between length of two trains is 3 : 2 and speed of both trains is 72 km/hr & 90 km/hr respectively. Faster train crosses slower trains in 60 sec running in same direction. If slower train crosses a goods train, which is running at the speed of 108 km/hr in 7.2 sec running in opposite direction. Then find the time taken by faster train to cross goods train running in same direction?

- (a) 40 sec
- (b) 64 sec
- (c) 60 sec
- (d) 56 sec
- (e) 50 sec

**Answer (c) 60 sec**

**342.** The speed of car is 25% more than the speed of bus. The time difference between them to cover a certain distance D, is 1 hour. On particular day, driver noticed that if they are (D- 40) km apart from each other then they can meet in 2 hours driving in opposite direction at their normal speed. Find 150% of the speed of bus ?

- (a) 150 km/hr
- (b) 90 km/hr
- (c) 120 km/hr
- (d) 105 km/hr
- (e) None of these

**Answer (c) 120 km/hr**

**343.** Time taken by car A to cover distance from point X to point Y is  $6\frac{1}{2}$  hrs including  $\frac{1}{2}$  hour of stoppage. B also started from point X and stopped 90 km earlier from point Y after travelling for  $6\frac{1}{2}$  hrs. Had they started from same point and at same time, the difference between their distance covered in 7.5 minutes will be 2.5 km. Find the distance between two cars, if they start a race for the distance which is twice the distance between X and Y , assuming that both cars stops when one of the car finish the race

- (a) 360 km
- (b) 240 km
- (c) 180 km

- (d) 300 km
- (e) 280 km

**Answer (b) 240 km**

**344.** Two trains A and B with their length difference 17 m cross each other in 2.6 seconds while running in opposite direction. Train A crosses train B in 13 seconds while they are running in same direction. If train A takes 2 seconds to cross a pole, then find the sum of speed of both the trains, assuming that train B is longer than train A.

- (a)  $34 \text{ ms}^{-1}$
- (b)  $51 \text{ ms}^{-1}$
- (c)  $68 \text{ ms}^{-1}$
- (d)  $85 \text{ ms}^{-1}$
- (e)  $102 \text{ ms}^{-1}$

**Answer (d)  $85 \text{ ms}^{-1}$**

**345.** 'Rajdhani express', 240 m long train crosses a pole in 6 sec and it also crosses two trains 'Shatabdi express' and 'Duranto express' in 17 sec & 42 sec respectively, where 'Shatabdi express' is running in opposite direction of 'Rajdhani express' and 'Duranto express' running in same direction as that of 'Rajdhani express'. If length of 'Shatabdi express' & 'Duranto express' is 360 m and 180 m respectively, then find in what time 'Shatabdi express' will cross 'Duranto express' while running in same direction?

- (a) 24 sec
- (b) 28 sec
- (c) 32 sec
- (d) 36 sec
- (e) 48 sec

**Answer (d) 36 sec**

**DIRECTIONS (Q. 346-347)**

There are two trains A and B, both train starts travelling to each other from stations P and Q respectively. Train A reaches the station Q in total 4 hours while train B reaches the station P in total 4 hours 48 minutes. Speed of train A is 120 km/hours.

**346.** Train A and B running on two different railroads parallelly in opposite direction to two different trains C and D respectively. Train A passed train C, which running at the speed of 60 km/hr. in 6 sec. and train B passed train D, which running at the speed of 110 km/hr. in 9 sec. if ratio between length of train A to train C is 3: 2, and that of between train B to train D is 4 : 3. Find in what time train D passed train C, if both running in same direction ?(2 marks)

- (a) 22.64 sec
- (b) 24.84 sec
- (c) 24.24 sec
- (d) 28.84 sec
- (e) 20.84 sec

**Answer (b) 24.84 sec**

**347.** Ratio between speed of train A to train E is 6: 5. Train E starts a journey from Kolkata to Delhi, which is 600 km at 9.45pm and train A starts same journey after 30 minutes of train E. Train A stop for x minutes on each three stoppages and train E stop for (x+10) minutes each on three stoppages between Kolkata to Delhi. Train A reaches Delhi one hour earlier than train E, which reaches at 4.30 am. Find in what time train B a total distance of 720 km cover between Lucknow to Jaipur. If train B stop for (x+10) minutes on four stoppages between Lucknow to Jaipur?(2 marks)

- (a) 4 hours
- (b) 8 hours
- (c) 8.2 hours

- (d) 10 hours
- (e) 12 hours

**Answer (c) 8.2 hours**

**348.** Speed of current is 10 km/hr and speed of a motor boat is 80% more than speed of current. Motor boat travels 280 km downstream with its usual speed, after that it's increased speed by 's' kmph and travelled for another 280 km then it returns and covers 560 km in upstream. If boat complete whole journey downstream to upstream in 45 hr, then find the value of 's'?

- (a) 10 km/hr
- (b) 8 km/hr
- (c) 6 km/hr
- (d) 12 km/hr
- (e) 4 km/hr

**Answer (d) 12 km/hr**

**349.** Speed of current is 5 km/hr and speed of boat in still water is 80% more than speed of current. A boat travels 50% of total distance of 280 km, in downstream. After that the speed increase by x kmph because of a man who added an engine. If boat complete whole journey downstream and upstream in 45 hr, then find the value of x?

- (a) 6 km/hr
- (b) 8 km/hr
- (c) 9 km/hr
- (d) 4 km/hr
- (e) 5 km/hr

**Answer (a) 6 km/hr**

**350.** Usual speed of a boat is S km/hr. Boat rowing in a lake (neglect the current speed) and covered distance of 360 km in three equal parts with three different speeds in total 7.5 hours. If boat covered first part with usual speed and second part with 20% increased speed and third part with further increase of 25% of latest speed. Find the speed of boat at which it covered third part of journey?

- (a) 56km/hr
- (b) 36km/hr
- (c) 40km/hr
- (d) 60km/hr
- (e) 48 km/hr

**Answer (d) 60km/hr**

**351.** When two ships A & B are travelling in opposite direction crosses each other in 2 second. Speed of ship B is 50% more than the speed of ship A in still water. On a particular day, ship A was ahead of ship B by 9 meters and both of them were travelling in downstream. If Ship B overtook Ship A in 11 seconds and length of ship B is 54 meters then find the speed of ship A.

- (a) 56m/s
- (b) 36m/s
- (c) 40m/s
- (d) 60m/s
- (e) 18m/s

**Answer (e) 18m/s**

**352.** A boat goes a certain distance downstream and then returns and covers 40% of distance covered in downstream. Ratio of time taken in covering downstream and upstream distances is 3 : 2. If speed of boat in still water is reduced by 50% then it covers 60 km downstream in 10 hours. Find the speed of boat in still water.

- (a) 9 km/hr
- (b) 8 km/hr
- (c) 6 km/hr
- (d) 10 km/hr
- (e) 12 km/hr

**Answer (b) 8 km/hr**

**353.** Circumference of a circular garden is 66 cm and area of circular garden is 25% more than perimeter of a square hall. Find the area of square hall ?

- (a) 4802.49 sq cm
- (b) 4704.48 sq. cm
- (c) 4820.49 sq.cm
- (d) 4822.49 sq.cm
- (e) none of these

**Answer (a) 4802.49 sq cm**

**354.** Internal radius of pipe is 3 cm and the external radius of pipe is 5 cm and the length of pipe is 15 cm then find total surface area of pipe ?

- (a)  $278 \pi \text{ cm}^2$
- (b)  $275 \pi \text{ cm}^2$
- (c)  $268\pi \text{ cm}^2$
- (d)  $272 \pi \text{ cm}^2$
- (e) none of these

**Answer (d)  $272 \pi \text{ cm}^2$**

**355.** A solid cube is melted to form 3 spheres, with radius, 1 cm, 2 cm and 3 cm and additional irregular shape was formed. Calculate the volume of irregular shape (upto two decimal place) if diagonal of cube is  $6\sqrt{3}$  cm.

- (a) 65.14 cm<sup>3</sup>
- (b) 70.35 cm<sup>3</sup>
- (c) 54.75 cm<sup>3</sup>
- (d) 50.00 cm<sup>3</sup>
- (e) 78.90 cm<sup>3</sup>

**Answer (a) 65.14 cm<sup>3</sup>**

**356.** 20% increase in both radius and height of the cylinder increases the total surface area of cylinder by 677.6 cm<sup>2</sup>, If the ratio of radius to height is 1 : 4, then find the radius of cylinder

- (a) 21 cm
- (b) 10.5 cm
- (c) 3.5 cm
- (d) 14 cm
- (e) 7 cm

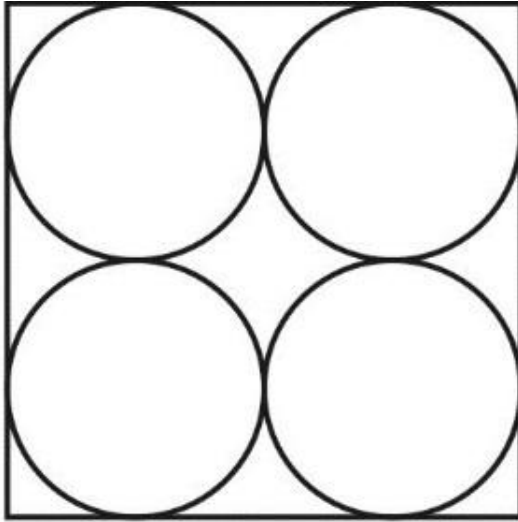
**Answer (e) 7 cm**

**357.** A Cone cylinder and hemisphere have equal radius and height. Find the ratio of total surface area of cylinder, cone and hemisphere.

- (a)  $4 : \sqrt{2} : \sqrt{6}$
- (b)  $4 : \sqrt{2} : 3$
- (c)  $\sqrt{2} + 1 : 4 : 3$
- (d)  $3 : 4 : \sqrt{2} + 1$
- (e)  $4 : \sqrt{2} + 1 : 3$

**Answer (e)  $4 : \sqrt{2} + 1 : 3$**

- 358.** Four identical coins are placed in a square. For each coin, area of each coin is equal to circumference of each coin. Then, find the area of the square that is not covered by the coins



- (a)  $16(\pi - 1)$
- (b)  $16(8 - \pi)$
- (c)  $16(4 - \pi) \pi$
- (d)  $16(4 - 2)$
- (e) None of these

**Answer (c)  $16(4 - \pi) \pi$**

- 359.** The surface area of a sphere is  $423.5 \text{ cm}^2$  less than total surface area of a hemisphere. If ratio between radius of hemisphere and sphere is  $3 : 2$ , then find the radius of hemisphere?

- (a) 5.5 cm
- (b) 5 cm
- (c) 4 cm
- (d) 7 cm
- (e) 10.5 cm

**Answer (e) 10.5 cm**

- 360.** The dimensions of a field are 20 m and 9 m. A pit 10 m long, 4.5 m wide and 3 m deep is dug in one corner of the field and the earth (mud) removed has been evenly spread over the remaining area of the field. What will be the rise the height of field as a result of this operation ?

- (a) 1.5 m
- (b) 2 m
- (c) 3 m
- (d) 4m
- (e) 1 m

**Answer (e) 1 m**

- 361.** Difference between perimeter of two rectangles is 8 cm. If breadth of both rectangle is 16 cm and length are in the ratio of  $7 : 6$  then find the total surface area of cylinder whose radius is half of the length of larger rectangle and height equal to length of smaller rectangle?

- (a)  $3344 \text{ cm}^2$
- (b)  $3668 \text{ cm}^2$
- (c)  $3444 \text{ cm}^2$
- (d)  $3468 \text{ cm}^2$
- (e)  $3462 \text{ cm}^2$

**Answer (a)  $3344 \text{ cm}^2$**

**DIRECTIONS (Q. 362-363)**

; There are three toys which are in cylindrical, conical and spherical shape. Height of all the toys is same. Ratio between the height of cylinder to radius of cylinder is 4 : 3.

**362.** Find the ratio between total surface area of cylinder to that of sphere.

- (a) 17 : 15
- (b) 21 : 16
- (c) 28 : 23
- (d) 21 : 8
- (e) 15 : 4

**Answer (d) 21 : 8**

**363.** Volume of cone is what percent of the volume of sphere if radius of cone is equal to that of cylinder

- (a) 112.5%
- (b) 100%
- (c) 125%
- (d) 75%
- (e) 88 9%

**Answer (a) 112.5%**

**364.** A well whose shape is of equilateral triangle having side 8 cm has to be dug  $44\sqrt{3}$  cm deep. The earth taken out of the well is used to fill a cylindrical tank having diameter 8 cm. Find the height up to which earth is filled in the tank.

- (a) 10.5 cm
- (b) 21 cm
- (c) 42 cm
- (d) 84 cm
- (e) 32 cm

**Answer (c) 42 cm**

**365.** A circle is inscribed in a square. If the difference between area of the square and circle is  $262.5 \text{ cm}^2$ , then find the area of the rectangle whose perimeter is same as that of circle while length of rectangle is 20% more than the breadth of rectangle (in  $\text{cm}^2$ )

- (a) 1500
- (b) 1400
- (c) 700
- (d) 750
- (e) 3000

**Answer (d) 750**

**366.** Inside a square plot a circular garden is developed which exactly fits in the square plot and the diameter of the garden is equal to the side of the square plot which is 28 metre. What is the area of the space left out in the square plot after developing the garden?

- (a) 98  $\text{m}^2$
- (b) 146  $\text{m}^2$
- (c) 84  $\text{m}^2$
- (d) 168  $\text{m}^2$
- (e) 68  $\text{m}^2$

**Answer (d) 168  $\text{m}^2$**



**367.** The area of a rectangle gets reduced by  $9\text{m}^2$  if its length is reduced by 5 m and breadth is increased by 3m. If we increase the length by 3 m and breadth by 2 m, the area is increased by  $67\text{m}^2$ . The length of the rectangle is :

- (a) 9 m
- (b) 15.6 m
- (c) 17 m
- (d) 18.5 m
- (e) 19 m

**Answer (c) 17 m**

**368.** A toymaker makes a toy in which a cone is mounted on the base of a hemisphere. If the total surface area of the toy is  $858\text{ cm}^2$  then find the volume of the toy, given that the diameter of the toy is 14 cm? 2 2

- (a)  $1950\sqrt{3}\text{ cm}^3$
- (b)  $1250\sqrt{3}\text{ cm}^3$
- (c)  $1400\text{ cm}^3$
- (d)  $1500\text{ cm}^3$
- (e)  $1200\text{ cm}^3$

**Answer (a)  $1950\sqrt{3}\text{ cm}^3$**

**369.** If sum of circumference of a circle and perimeter of rectangle is 204 cm and perimeter of rectangle is 50% of magnitude of area of square and side of square is 12 cm .If length of rectangle is 8 cm more than that of breadth, then find sum of area of circle and rectangle ?

- (a)  $1694\text{sq cm}$
- (b)  $1450\text{sq cm}$
- (c)  $1550\text{sq cm}$
- (d)  $1750\text{sq cm}$
- (e)  $1460\text{sq cm}$

**Answer (a)  $1694\text{sq cm}$**

**370.** A sphere is melted into 'n' number of small spheres. Total surface area of these small spheres is what percent more or less than total surface area of large sphere, if the ratio of radius of small sphere to radius of large sphere is 1 : 3.

- (a) 50%
- (b) 100%
- (c) 150%
- (d) 200%
- (e) 75%

**Answer (d) 200%**

**371.** Ratio of height of cylinder to that of diameter is 2: 3. This cylinder is formed by melting a sphere having same radius as the radius of circle. Ratio of magnitude of area of circle to magnitude of its circumference is 21: 2. Find height of cylinder.

- (a) 14 units
- (b) 21 units
- (c) 42 units
- (d) 28 units
- (e) 35 units

**Answer (d) 28 units**

**372.** If the curved surface area of first cone is thrice that of second cone and slant height of the second cone is thrice that of the first, find the ratio of the area of their base.

- (a) 81 : 1
- (b) 9 : 1
- (c) 3 : 1
- (d) 27 : 1
- (e) None of these

**Answer (a) 81 : 1**

**373.** There are two cylindrical vessels with same capacity and same dimensions. If radius of one vessel increased by 7 cm and height of second vessel increases by 15 cm, then capacity of both vessel increased by equality 'Q' cm<sup>3</sup>. If actual height of both vessels is 12 cm, then find value of Q?

- (a) 9840 cm<sup>3</sup>
- (b) 9240 cm<sup>3</sup>
- (c) 9460 cm<sup>3</sup>
- (d) 9160 cm<sup>3</sup>
- (e) 9780 cm<sup>3</sup>

**Answer (b) 9240 cm<sup>3</sup>**

**374.** There are four light poles in four corners of a park, four at midpoint of each side and one pole in the middle of that square park. Area of park is 19600 cm<sup>2</sup>. If side of park is 50 times of radius of each light pole, then find the total base area covered by these poles if pole's base is circular in shape?

- (a) 197.12 cm<sup>2</sup>
- (b) 172.48 cm<sup>2</sup>
- (c) 246.4 cm<sup>2</sup>
- (d) 220.76 cm<sup>2</sup>
- (e) 221.76 cm<sup>2</sup> 9

**Answer (e) 221.76 cm<sup>2</sup> 9**

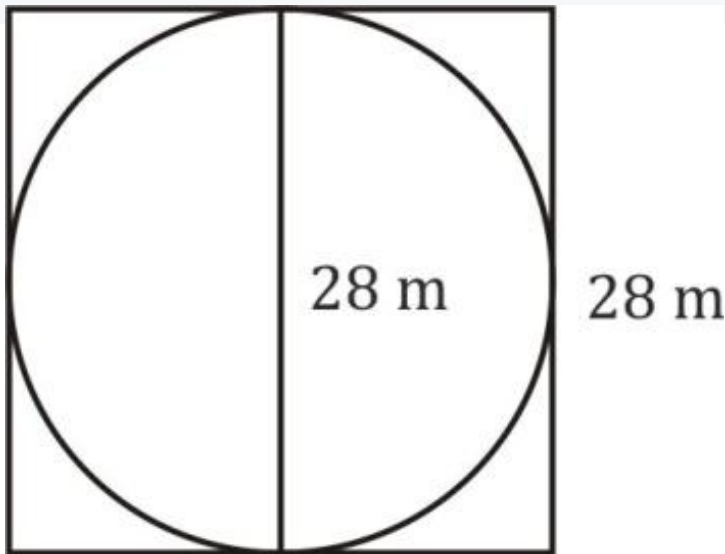
**375.** Area of a square is 306.25 cm<sup>2</sup> and breadth of rectangle is 3.5 cm less than side of square. Area of rectangle is 252 cm<sup>2</sup> and height and radius of a cylindrical vessel equal to length & breadth of rectangle respectively. If vessel filled, contains mixture of milk and water in which milk is 75%, then find the volume of water contained by cylindrical vessel?

- (a) 2572 cm<sup>3</sup>
- (b) 2472 cm<sup>3</sup>
- (c) 2772 cm<sup>3</sup>
- (d) 2872 cm<sup>2</sup>
- (e) 2878 cm<sup>2</sup>

**Answer (c) 2772 cm<sup>3</sup>**

**DIRECTIONS (Q. 376-377)**

; A hemispherical bowl is filled with hot water to the brim. The contents of the bowl are transferred into a cylindrical vessel whose radius is 50% more than its height.



**376.** If diameter of the bowl is the same as that of the cylindrical vessel, then the volume of the hot water in the cylindrical vessel is

- (a) 60% of the cylindrical vessel
- (b) 80% of the cylindrical vessel
- (c) 100% of the cylindrical vessel
- (d) 50% of the cylindrical vessel
- (e) none of these

**Answer (c) 100% of the cylindrical vessel**

**377.** This hemispherical bowl is joined at one end of the cylindrical vessel and the solid obtained by combining these figure is filled completely with water. If a drain pipe is connected to it then pipe will empty  $539 \text{ cm}^3$  of water in one minute. If radius of the solid is 21 cm, then in what time whole of the water will be emptied from the solid. Given that diameter of cylinder and hemisphere are same.

- (a) 72 min
- (b) 60 min
- (c) 90 min
- (d) 75 min
- (e) None of these

**Answer (a) 72 min**

**378.** A cylinder whose height is equal to its radius is full of milk. Its milk is poured into a hemi spherical bowl of same radius as of cylinder and remaining milk is poured into another hemi-spherical bowl of same volume as of previous one. What percentage of the volume of second hemispherical bowl remained empty?

- (a) 40%
- (b)  $66\frac{2}{3}\%$
- (c)  $33\frac{1}{3}\%$
- (d) 50%
- (e) 100%

**Answer (d) 50%**

**379.** In a house, there is a semicircular corridor, a circular hall and a circular garden. Sum of perimeter of corridor and hall is 102 m. If radius of hall is 50% more than radius of corridor then find area of circular garden, given radius of garden is 300% more than sum of radius of corridor and radius of hall together.

- (a)  $6300\pi$  sq.m
- (b)  $5400\pi$  sq.m
- (c)  $5700\pi$  sq.m
- (d)  $4200\pi$  sq.m
- (e)  $4900\pi$  sq.m

**Answer (e)  $4900\pi$  sq.m**

**380.** Height of a cylindrical vessel is equal to side of a square, whose area is  $256 \text{ cm}^2$ . Radius of cylindrical vessel is equal to average of radius of a sphere and hemisphere. If ratio between the radius of hemisphere and that of the sphere is 2 : 1 and difference between their total surface area is  $4928 \text{ cm}^2$ , then find volume of cylindrical vessel ?

- (a)  $21,288 \text{ cm}^3$
- (b)  $20,176 \text{ cm}^3$
- (c)  $22,176 \text{ cm}^3$
- (d)  $26,176 \text{ cm}^3$
- (e)  $24,288 \text{ cm}^3$

**Answer (c)  $22,176 \text{ cm}^3$**

**381.** Radius of a cylinder is half of the side of square, whose perimeter is 6 cm more than perimeter of rectangle. If ratio between length to breadth of rectangle is 7 : 2 and between length of rectangle to side of square is 3 : 2. Find volume of cylinder, given height of cylinder is equal to breadth of rectangle?

- (a)  $24,938 \text{ cm}^3$
- (b)  $24,948 \text{ cm}^3$
- (c)  $24,848 \text{ cm}^3$
- (d)  $24,996 \text{ cm}^3$
- (e)  $24,918 \text{ cm}^3$

**Answer (b)  $24,948 \text{ cm}^3$**

**382.** Radius of a cylindrical vessel is 1.5 cm less than breadth of rectangle, while height of cylindrical vessel is equal to length of rectangle, whose area is  $168 \text{ cm}^2$ . If ratio between length & breadth of rectangle is 7 : 6, then find the thickness of 1 disc, if these discs are kept one above another in cylindrical vessel and 10 discs get completely fit in cylindrical vessel. (Let radius of discs equal to radius of cylindrical vessel)?

- (a) 1.2 cm
- (b) 1.6 cm
- (c) 1.8 cm
- (d) 1.4 cm
- (e) 1.0 cm

**Answer (d) 1.4 cm**

**383.** The ratio between radius of two hemispheres solid tin pieces is 2 : 3 and difference between volume of both is  $\frac{836}{21} \text{ cm}^3$ . These two  $\pi$  hemispheres are melted into a cylindrical vessel and used  $74 \frac{3}{4} \text{ cm}^3$  extra tin material for polishing the vessel. If ratio between height & radius of cylindrical vessel is 3 : 4, then find the total surface area of cylindrical vessel?

- (a)  $154 \text{ cm}^2$
- (b)  $132 \text{ cm}^2$
- (c)  $176 \text{ cm}^2$
- (d)  $208 \text{ cm}^2$
- (e)  $198 \text{ cm}^2$

**Answer (c) 176 cm<sup>2</sup>**

**384.** Sum of digits of a 5-digit number is 43. Find the probability that such a number is divisible by 11?

- (a)  $\frac{1}{3}$
- (b)  $\frac{1}{5}$
- (d)  $\frac{2}{15}$
- (e)  $\frac{2}{3}$
- (c) 66 3 5 1 2 3 3, 5 and 7 respectively. 29 19 105 35 1 5 5 7 16 16 9 16 6 4 11 11 16 33 Publications@

**Answer (b)  $\frac{1}{5}$**

**385.** There are three groups of children in a school i.e. A, B and C. In group A, there are  $(x + 2)$  girls and  $x$  boy. In group B, there are 2 boys and 2 girls, whereas in group C, there are 1 girl and 2 boys respectively. One child is selected at random from each group. Find the value of  $x$ , if the probability of selecting 1 girl and 2 boys is

- (a) 2
- (b) 3
- (d) 4
- (e) 5
- (c) 120 7 23 6 4 11 11 16 33 5

**Answer (c) 120 7 23 6 4 11 11 16 33 5**

**DIRECTIONS (Q. 386)**

There are some horses, hens and elephants in a park. Ratio between number of heads in park to number of legs in park are 7 : 20.

**386.** If probability of choosing 2 horses over all the animals are the number of elephant present in park. (Number of horses is double than that of the number of elephants in park).

- (a) 5
- (b) 10
- (d) 15
- (e) 25
- (c) 31 5

**Answer (a) 5**

**DIRECTIONS (Q. 387-388)**

In a stationary shop there are four types of colored sheets of red, blue, green and white colors. The probability of selecting one 1 red sheet out of the total sheets is  $\frac{3}{5}$ , the probability of selecting one blue 2 sheet out of the total sheets is  $\frac{7}{10}$  and the probability of selecting one white 1 sheet out of the total sheets is  $\frac{1}{10}$

**387.** If all the sheets are numbered as 1,2,3,...and so on and one sheet is picked up at random, then find the probability of picking up a sheet which is numbered as a multiple of 13 or 17. (2 marks) 1 1 5

- (a) 8
- (b) 7
- (c) 8 9
- (d) 10
- (e) none of these

**Answer (a) 8**

**388.** What is the total number of sheets in the bag? (1 marks)

- (a) 117
- (b) 168
- (c) 154
- (d) 120
- (e) none of these

**Answer (b) 168**

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**DIRECTIONS (Q. 389-390)**

A coach invited 9 players of the team on dinner. All 10 persons seated in a circular table.

**389.** In how many ways captain and vice-captain seat opposite to each other?

- (a) 80640
- (b) 5040
- (c) 3624
- (d) 40320
- (e) 10080

**Answer (e) 10080**

**390.** In how many ways team can seat, if caption and vice-captain seat either side of coach.

- (a) 80640
- (b) 5040
- (c) 3624
- (d) 40320
- (e) 10080

**Answer (e) 10080**

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**DIRECTIONS (Q. 391)**

A joker have 10 balls, 10 ribbon and 10 plates of red, yellow and blue color in ratio 2 : 3 : 5, 5 : 2 : 3 and 3 : 5 : 2 respectively. He does his performance by using one item of each category.

**391.** What is the probability that he performs with only red color?

- (a) 0.3
- (b) 0.02
- (c) 0.25
- (d) 0.03
- (e) 0.15 12

**Answer (d) 0.03**

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**392.** I.  $3x+5.92x-4 = 95x-14$

II.  $2y^2 - 15y - 28 = 3y^2 - 23y - 13$

- (a)  $x < y$
- (b)  $x \geq y$
- (c) No relation can be established between x and y
- (d)  $x \leq y$
- (e)  $x > y$

**Answer (a)  $x < y$**

393. I.  $x^2 - x + 2 = 0$  40 13

II.  $y^2 + 1 = y$

- (a)  $x < y$
- (b)  $x \geq y$
- (c) No relation can be established between  $x$  and  $y$
- (d)  $x \leq y$
- (e)  $x > y$

**Answer (b)  $x \geq y$**

394. I.  $2(y+4)$ .  $12(2y+4) = 9(y+2)$ .  $16(y+4)$

II.  $x^2 - 10x + 24 = 0$

- (a)  $x \leq y$
- (b)  $x > y$
- (c) No relation can be established between  $x$  and  $y$
- (d)  $x < y$
- (e)  $x \geq y$  48 14

**Answer (b)  $x > y$**